

ENGINEERING DESIGN HANDBOOK

VCH Method

A Sizing Framework for Submerged Hydronic Systems

PREPARED BY

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First Edition

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The development of the Velocity-Constrained Hydronic (VCH) Sizing Framework represents a deeply challenging personal and technical journey. While the pages of this handbook contain rigorous fluid dynamics and mechatronic control equations, its completion was made possible through a unique combination of industry inspiration, familial support, and modern computational tools.

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— **Sambridda Ranabhat** *Kathmandu, Nepal*

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Abstract

This document presents the Velocity-Constrained Hydronic (VCH) Sizing Framework — a structured, iterative engineering methodology for the thermal and hydraulic design of submerged hydronic heating and cooling systems used in bulk liquid thermal processing applications. The framework addresses the fundamental inadequacy of linear design approaches by integrating thermal demand analysis, thermofluid selection, vessel geometry, hydraulic resistance characterisation, heat transfer evaluation, coil optimisation, environmental loss quantification, thermal driver sizing, and thermal battery design into a unified eleven-step engineering workflow culminating in a formal design verification gate.

The methodology introduces two distinct operating mode demands — the batch processing target, which accounts for the full sensible and latent energy required to process incoming cold raw material within a defined operational window, and the balance target, which quantifies the steady-state energy required to maintain a vessel already at operating temperature against environmental heat loss. The governing demand is established as an independent fixed design target before any hardware is selected, eliminating the circular dependency inherent in frameworks that derive their design target from the hardware they are sizing.

Practical engineering constraints are embedded throughout the framework, including velocity-based erosion ceilings enforced against commercially available Nominal Pipe Sizes, a compound coil packing efficiency correction accounting for fabrication tolerances, structural support requirements, weld access, thermal expansion, and dead-zone prevention, and a worst-case ambient temperature reference applied consistently across all thermal boundary calculations. The framework incorporates a rigorous series thermal resistance network for environmental heat loss quantification, Effectiveness-NTU heat transfer evaluation with separate steady-state and cold-start coefficient assessments, and a thermal battery sizing procedure that produces a complete custom tank geometry bounded by hydraulic priming and driver stability constraints.

The resulting methodology provides system designers, mechanical engineers, and mechatronics integrators with a standardised, auditable, and physically grounded design protocol that is inherently safeguarded against long-term fouling degradation, hydraulic instability, and thermal underperformance.

Introduction

The thermal management of bulk liquid storage and processing vessels is a foundational requirement across the chemical, food processing, agricultural commodity, and heavy manufacturing industries. The engineering challenge appears deceptively straightforward: heat a volume of fluid from one temperature to another within a defined time. In practice, however, the design of the internal hydronic system that accomplishes this task is governed by a tightly coupled set of thermal, hydraulic, geometric, and control constraints that interact in ways a simple linear calculation cannot capture.

The traditional approach to hydronic coil sizing treats the problem as a sequential thermodynamic exercise. The designer calculates the thermal load, determines the required heat exchange surface area, selects a pipe size, and assembles the coil. Each step is performed in isolation. The result is a design that is thermodynamically plausible on paper but routinely fails in practice for reasons that are entirely predictable. Pipe diameters derived from unconstrained calculations do not correspond to commercially available sizes. Coils sized to theoretical thermal minimums degrade rapidly under organic fouling and provide no cold-start surge capacity. Vessels fabricated to theoretical volumes deliver less usable process capacity than specified once the coil assembly is physically installed. Pumps selected without reference to velocity limits drive the fluid at erosive speeds that destroy the coil from within.

These failures are not the result of calculation errors. They are the result of a design philosophy that evaluates mechanical constraints in isolation rather than as an integrated system. The engineer who calculates thermal surface area without simultaneously constraining fluid velocity, pipe diameter, and coil packability will produce a design that is correct in theory and unworkable in practice.

The Velocity-Constrained Hydronic (VCH) Sizing Framework was developed to resolve this gap. The methodology emerged from the design of a real industrial thermal circulation system involving submerged hydronic coils for viscous fluid heating, where the inadequacy of conventional sizing tools became apparent during the translation from theoretical design to physical fabrication. The framework was built to capture the lessons of that translation and encode them into a repeatable, structured engineering process.

The VCH framework departs from the linear approach in three fundamental ways. First, the thermal design target is established independently of the hardware before any physical component is selected. The required process heat load — accounting for batch processing demands and steady-state holding losses under worst-case ambient conditions — is fixed at the outset and multiplied by a design factor of safety. Every subsequent hardware decision is evaluated against this fixed target. It is never re-derived from the system's own capacity.

Second, all pipe geometry is constrained to commercially available Nominal Pipe Sizes from the outset. Theoretical internal diameters are not permitted. The velocity ceiling is assigned before the coil is designed and governs every hydraulic and automation decision that follows.

Third, the framework evaluates the system's performance under two distinct thermal conditions — steady-state bulk operating conditions and worst-case cold-start conditions. A system that meets its thermal target only at

stabilised operating temperature but fails during startup is not an adequate design. The VCH framework sizes to the more demanding of the two conditions in every case.

The result is a design methodology that produces thermal management assets which are structurally sound, commercially manufacturable, hydraulically stable, thermally resilient, and inherently optimised for the closed-loop automation systems that govern modern industrial plant operations.

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This framework is provided in the spirit of open engineering collaboration and the advancement of human knowledge.

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Scope and Operational Boundaries

To preserve the mathematical validity, physical safety, and operational predictability of the VCH Sizing Framework, its engineering boundaries must be precisely defined. Applying the framework outside its validated boundary conditions introduces risks of mechanical failure, control instability, and capital loss. This section establishes the explicit technical inclusions and exclusions that govern the deployment of the VCH method.

Intended Applications

The VCH Sizing Framework is explicitly compiled, mathematically modelled, and physically validated for systems operating within the following technical domains:

Submerged Coil Liquid-to-Liquid Heat Exchangers — Designed specifically for internal helical or serpentine coil configurations completely submerged within a stationary or low-velocity bulk process fluid contained in a cylindrical vessel with conical or hemi-ellipsoidal caps.

Single-Phase Closed-Loop Hydronic Systems — Tailored for circulation systems utilising single-phase liquid thermal transport media — including treated water, ethylene glycol blends, synthetic thermal oils, and food-grade heat transfer fluids — operating below their boiling points and above their freezing thresholds at system operating pressure.

Sensible Process Heating and Cooling — Validated for applications where the primary thermal objective is a controlled temperature elevation or maintenance of process fluids through sensible heat transfer. The framework accommodates phase-change energy contributions in the incoming batch material — such as the melting of solid feedstocks — as a component of the batch thermal demand calculation, but does not model phase change within the thermofluid itself.

High-Viscosity Fluid Thermal Processing — Optimised for the non-linear temperature-viscosity behaviour characteristic of heavy organic products such as edible plant oils, heavy hydrocarbons, waxes, and liquid-phase polymers. The framework explicitly evaluates heat transfer performance under worst-case cold-start viscosity conditions for this reason.

Multi-Vessel Industrial Plant Layouts — Capable of sizing a centralised thermal driver and thermal battery to serve multiple parallel process vessels simultaneously, with total capacity governed by simultaneous peak loading across all active nodes.

Unintended Applications

The foundational assumptions of the VCH framework — specifically the steady-state single-phase Darcy-Weisbach hydraulic friction model and the Effectiveness-NTU thermodynamic equations — render it invalid for the following equipment classifications:

Phase-Change Generation Systems — The framework is not intended for steam boilers, flash evaporators, or any thermal asset where the utility fluid undergoes a liquid-to-vapour phase transformation. Such systems require specialised vapour-fraction velocity limits and latent heat transport formulations.

Cryogenic and Deep-Refrigeration Systems — The material selection matrices, fouling factor allowances, and thermal expansion parameters are structurally unsuited to cryogenic regimes where localised structural embrittlement and extreme fluid density shifts dominate.

Multiphase Transport Regimes — The velocity ceiling and pipe sizing rules do not apply to slug, plug, or annular multiphase flows where erosion mechanics are governed by particulate impact rather than clear fluid shear kinetics.

Direct-Fired and Turbulent Combustion Systems — This method is entirely separated from direct-fired thermal oxidisers, radiant furnace tubes, or combustion-driven heat recovery configurations operating under radiation-dominated heat flux profiles.

Open-Loop Systems — The hydraulic and thermal battery sizing formulations assume a closed recirculating loop with a fixed fluid inventory. Open-loop systems where fluid is consumed or discharged rather than recirculated are outside the validated scope of this framework.

Note on Cooling Applications

The thermodynamic and hydraulic principles underlying this framework are applicable to cooling loops with symmetric logic. Where the direction of constraint, fouling behaviour, or control safety logic differs for cooling configurations — specifically the reversal of the worst-case ambient reference, the risk of accidental process heating if the chiller loop temperature rises above the tank temperature, and the distinct minimum volume constraints of vapor compression equipment — these differences are identified explicitly within the relevant steps. The framework is oriented primarily toward industrial heating. Cooling applications may be designed using this methodology with the noted modifications applied conscientiously.

Governing Assumptions

To establish analytical tractability and ensure a mathematically rigorous foundation, the VCH framework relies on a set of standardised engineering assumptions. Explicitly stating these assumptions delineates the physics accounted for by the core design equations and isolates the variables managed through structural safety factors and empirical correction coefficients.

Steady-State Operational Baseline The primary hydraulic and thermal formulations are evaluated under time-independent, steady-state continuous operating conditions. Transient thermal behaviour — including the cold-startup phase and batch loading transients — is managed through the application of the design factor of safety, the worst-case cold-start heat transfer coefficient, and the thermal battery volume selection. The baseline sizing calculations assume that fluid velocities, mass flow rates, and boundary temperatures have stabilised.

Incompressible Fluid Flow The thermofluid traversing the internal coil network is assumed to be an incompressible liquid operating entirely in single liquid phase safely below its vapour pressure ceiling. This validates the direct application of the continuity equation and the Darcy-Weisbach friction loss model throughout the framework.

Newtonian Fluid Approximation Both the internal thermofluid and the external bulk process fluid are treated as Newtonian fluids within their active operating temperature bands. This assumption implies that viscous shear stresses are linearly proportional to the local strain rate, allowing the framework to utilise standard Reynolds number formulations and the Colebrook-White friction factor correlation.

Uniform Bulk Process Temperature The bulk process fluid contained within the primary storage vessel is assumed to maintain a uniform macro-scale bulk temperature at any distinct steady-state operating point. This assumes that natural buoyant convection currents or external mechanical agitation are sufficient to prevent severe localised thermal stratification within the tank core, allowing the external thermal resistance to be modelled using a localised convective boundary layer coefficient rather than a complex three-dimensional thermal gradient.

Quasistatic Piecewise Fluid Properties Thermophysical properties of the thermofluid — density, specific heat capacity, dynamic viscosity, and thermal conductivity — are treated as constant within a given calculation pass. These properties are extracted as piecewise values evaluated at the mean bulk operating temperature of the thermofluid within the circulation loop. The ΔT_{coil} constraint enforced throughout the framework limits the temperature variation of the thermofluid across the coil to ten percent of the operating-to-ambient range, bounding the property variation and preserving the validity of this assumption.

Single-Phase Hydronic Transport The fluid regime inside the internal heat exchange coil is strictly restricted to single-phase liquid transport. The framework assumes zero localised boiling, flashing, or vapour-fraction generation along the internal pipe walls.

Negligible Radiative Heat Losses External heat dissipation from the outer vessel surface is assumed to be dominated by conduction through the insulation layer and subsequent convection to the ambient air. Radiative heat transfer from properly cladded industrial insulation surfaces is minor and is mathematically consolidated into the external convective boundary layer resistance.

Known Worst-Case Ambient Temperature The framework assumes that the historically worst-case ambient temperature for the installation site — coldest recorded for heating applications, hottest recorded for cooling applications — is available to the designer at the outset of the design process. This value is applied consistently as the external thermal boundary condition across all steps where ambient temperature appears.

Available Process Fluid Thermophysical Properties The framework assumes that the thermophysical properties of the process fluid — specific heat capacity in solid and liquid phases, latent heat of fusion where applicable, density, viscosity, and thermal conductivity — are known or can be reliably estimated from published data or experimental measurement. For novel or proprietary process fluids where these properties are not available, the batch thermal demand calculations in Step 1 cannot be executed without supplementary characterisation work.

Glossary of Terms

The following terms are used throughout the VCH framework. They are defined here to ensure consistent interpretation across all engineering disciplines engaged with this document.

Batch Processing Mode The peak thermal demand operating state of a process vessel during which raw material is actively being heated from its incoming condition to the target operating temperature. This mode imposes the highest instantaneous thermal load on the system.

Balance Mode The steady-state minimum demand operating state of a process vessel during which the vessel is already at operating temperature and the system is maintaining thermal equilibrium against environmental heat loss only. No material is being actively processed.

Coil Packing Efficiency (η_{pack}) The compound dimensionless multiplier applied to the theoretical maximum coil length to account for real-world fabrication, structural, and operational constraints. It is the product of five sub-factors: fabrication quality, support structure, weld clearance, thermal expansion, and dead-zone prevention.

Dean Number (De) A dimensionless parameter quantifying the intensity of the secondary counter-rotating flow structures — Dean vortices — generated by fluid flowing through a curved or helical pipe. Defined as $De = Re \times \sqrt{(d_{in} / D_{coil})}$. Governs the selection of the appropriate helical friction factor and Nusselt number correlations.

Dean Vortices Secondary counter-rotating flow cells generated by the centrifugal force field acting on fluid traversing a helical pipe bend. These vortices increase the effective friction factor and enhance the internal convective heat transfer coefficient relative to straight pipe flow of identical geometry.

Design Factor of Safety (FOS) A dimensionless multiplier applied to the governing process heat load to establish the design thermal target. Accounts for long-term thermal performance degradation due to fouling, uncertainty in heat transfer coefficient estimation, manufacturing tolerance effects, and cold-start surge demand. Minimum value 1.5, recommended value 2.0, maximum recommended value 2.5.

Design Thermal Target (Q_{design}) The fixed independent thermal power benchmark against which all hardware decisions in the framework are evaluated. Defined as $Q_{required} \times FOS$. Established in Step 1 and carried forward unchanged through all subsequent steps.

Duty Point The steady-state hydraulic equilibrium condition at which the pump-generated head exactly equals the total system resistance. Defined by the intersection of the pump performance curve and the system resistance curve. Represents the natural unregulated operating condition of the circulation loop at a given pump speed.

Effectiveness (ϵ) The dimensionless ratio of the actual heat transfer rate to the thermodynamic maximum possible heat transfer rate. Bounded strictly between 0 and 1. Used in the Effectiveness-NTU method to characterise heat exchanger thermal performance.

Factor of Safety — see **Design Factor of Safety**

Fouling Resistance (R_f) The thermal resistance introduced by deposit layers accumulating on internal or external pipe surfaces over the operational service life. Applied to both the thermofluid-side and process fluid-side surfaces in the overall heat transfer coefficient calculation to ensure the coil meets its thermal target under degraded long-term operating conditions.

Head Loss The reduction in total hydraulic energy experienced by a fluid traversing a piping network, expressed as an equivalent height of fluid column. Represents the conversion of mechanical fluid energy into thermal energy via internal friction, wall shear stress, and localised turbulence at fittings and bends.

Hydronic Priming Volume ($V_{hydronic}$) The minimum thermofluid inventory required to completely fill all internal wetted passages of the closed circulation loop — including the coil bore, distribution pipework, and ancillary equipment — under operating pressures. The unconditional minimum volume floor for the thermal battery regardless of driver type.

Logarithmic Mean Temperature Difference ($LMTD / \Delta T_{lm}$) The effective thermal driving force between two fluid streams in a heat exchanger, accounting for the exponential change in temperature differential along the length of the exchange surface. For a submerged coil in a well-mixed vessel with uniform bulk process temperature, the LMTD is evaluated between the thermofluid inlet and outlet temperatures and the constant process fluid bulk temperature.

Maximum Practical Coil Length ($L_{practical}$) The maximum physically realisable coil length within the vessel envelope after applying the compound packing efficiency correction to the theoretical maximum. Represents the upper boundary of the system's heat exchange capacity given real fabrication and operational constraints.

Nominal Pipe Size (NPS) A standardised pipe designation system defined by ASME B36.10M and B36.19M specifying commercially available external diameters, wall thicknesses, and internal flow areas for welded and seamless pipe. All pipe geometry in the VCH framework is constrained to NPS-compliant dimensions.

Number of Transfer Units (NTU) A dimensionless parameter quantifying the thermal sizing of a heat exchanger relative to the fluid heat capacity rate. Higher NTU values indicate a physically larger heat exchanger approaching its thermodynamic limit. For $C_r = 0$, $NTU = -\ln(1 - \epsilon)$.

Overall Heat Transfer Coefficient (U) The composite thermal conductance of the complete heat exchange interface between the thermofluid and the process fluid per unit of external pipe surface area. Accounts for internal convection, pipe wall conduction, external convection, and fouling resistances on both surfaces. Evaluated at both steady-state and cold-start conditions.

Pump Affinity Laws The governing relationships between pump rotational speed and hydraulic performance. Flow rate is proportional to speed, head is proportional to speed squared, and power is proportional to speed cubed. Used to determine the VFD frequency required to clamp fluid velocity to the allowable ceiling.

$Q_{balance}$ The estimated continuous thermal power required to maintain a process vessel already at operating temperature against steady-state environmental heat loss. Represents the minimum sustained thermal demand of the installation during idle holding periods between active batch cycles.

Q_{batch} The continuous thermal power required to process a full incoming batch of raw material from worst-case ambient temperature to the target operating temperature within the specified operational window. Accounts for sensible heating of the solid phase, latent heat of fusion where applicable, and sensible heating of the liquid phase.

Q_{design} — **see Design Thermal Target**

Q_{loss} The steady-state rate of thermal energy dissipation from the process vessel to the surrounding environment through the vessel wall and insulation. Used as a placeholder estimate in Steps 1 through 8 and replaced by a rigorous thermal resistance network calculation in Step 9.

$Q_{required}$ The governing process heat load — the larger of Q_{batch} and $Q_{balance}$ — representing the peak thermal demand the system must reliably satisfy. $Q_{design} = Q_{required} \times FOS$.

Reynolds Number (Re) A dimensionless ratio comparing inertial forces to viscous forces within a moving fluid. The primary metric for determining whether a fluid regime is laminar, transitional, or turbulent. Governs the selection of friction factor correlations, Nusselt number correlations, and flow regime classifications throughout the framework. Defined as $Re = \frac{\rho \times v \times d_{in}}{\mu}$.

System Resistance Curve The relationship between volumetric flow rate and the total pressure loss required to drive fluid through the piping network at that flow rate. Composed of major friction losses and minor fitting losses. Characteristically parabolic — pressure loss increases approximately with the square of flow velocity.

Thermal Battery The custom-fabricated utility fluid reservoir that stores conditioned thermofluid and supplies it continuously to the circulation loop serving the process vessels. Sized between minimum volume constraints governed by hydraulic priming and driver stability, and a maximum volume constraint governed by peak batch autonomy.

Thermal Driver The primary commercial heating or cooling source — electric immersion heater assembly, fired thermal fluid heater, industrial heat pump, or vapor compression chiller — that conditions the thermofluid to the required supply temperature and delivers it to the thermal battery.

Thermal Time Constant (S_t) A stability metric quantifying the thermal inertia of the thermal battery relative to the peak process demand. Defined as $S_t = \rho \times V_{battery} \times C_p / Q_{peak}$. Expressed in seconds. Higher values indicate greater thermal stability and less demanding control hardware requirements.

Thermofluid The primary thermal transport medium circulating within the closed hydronic loop. Carries thermal energy from the thermal driver through the thermal battery and internal coil network to the process vessel.

Common classes include treated water, ethylene glycol-water solutions, synthetic thermal oils, and food-grade thermal oils.

Variable Frequency Drive (VFD) An electronic motor speed controller that regulates pump rotational speed by varying the frequency of the electrical supply to the pump motor. Used in the VCH framework to clamp the circulation pump operating speed — and therefore fluid velocity — to or below the allowable velocity ceiling.

Velocity Ceiling (v_{max}) The maximum allowable steady-state fluid velocity within the internal heat exchange coil. Determined by the combination of thermofluid class and pipe material. Enforced as a hard constraint on all pump selection, VFD configuration, and automation control logic throughout the framework. Exceeding this ceiling causes accelerated internal pipe erosion, flow-induced vibration, and cavitation.

Worst-Case Ambient Temperature ($T_{worst\ ambient}$) The historically most extreme ambient temperature recorded at the installation site. For heating applications, the coldest recorded ambient temperature. For cooling applications, the hottest. Applied consistently as the external thermal boundary condition across batch energy calculations, ΔT_{coil} constraints, heat loss calculations, and thermal battery maximum volume derivations.

ΔT_{coil} The temperature difference between the thermofluid entering and leaving the internal heat exchange coil. Capped at ten percent of $(T_{operating} - T_{worst\ ambient})$ to ensure thermofluid thermophysical properties remain sufficiently uniform for single-pass lumped-parameter calculations to remain valid.

ΔT_{drive} The temperature difference between the thermofluid inside the coil and the process fluid in the tank. The thermal driving force across the coil wall. Governs the rate of heat transfer through the exchange surface and is used in LMTD and Effectiveness-NTU calculations.

ΔT_{loss} The temperature difference between the process fluid at operating temperature and the worst-case ambient temperature. Drives the steady-state environmental heat loss through the vessel wall and insulation system.

How to Use This Document

This document is structured as a sequential engineering design guide. The eleven steps of the core methodology are designed to be executed in order. Each step produces a defined set of outputs that serve as inputs to the steps that follow. Skipping steps or executing them out of sequence will produce a design that is internally inconsistent and cannot be formally verified.

Document Structure

The document is organised into the following major sections:

Front Matter — Abstract, Introduction, Scope and Operational Boundaries, Governing Assumptions, Glossary of Terms, and this How to Use section. Read these sections in full before beginning the design process. The governing assumptions define the physical conditions under which the framework is valid. The scope section defines the applications for which it is not.

Framework Overview — A concise prose description of the eleven-step methodology with the three temperature differential definitions and the closed-loop iteration logic. Use this section to understand the overall structure of the design before engaging with the step-by-step detail.

Core Methodology — Steps 1 through 11 — The detailed engineering procedure. Each step follows a consistent structure: an overview explaining what the step accomplishes and why, numbered subsections covering each element of the calculation, an iteration gate or verification check where applicable, and a step output summary listing every value produced by the step. Each step concludes with a reference table identifying the governing equations derived in Appendix A.

Appendix A — Governing Mathematical Derivations — The full algebraic and numerical derivations of all governing equations used in the main body steps. Referenced from each step by equation identifier. Consult this appendix when the derivation basis of a governing equation is required for peer review, regulatory submission, or academic verification.

Appendix B — Thermophysical Fluid Property Matrices — Tabulated thermophysical properties for common thermofluid classes at discrete operating temperatures, with linear interpolation guidance for intermediate values.

Appendix C — Commercial Piping Standards and Material Profiles — ASME B36.10M and B36.19M pipe dimensional data and material evaluation matrices. Referenced during NPS selection in Step 4 and material confirmation in Step 4.

Appendix D — Distributed Process Automation and Control Architecture — The system-level control philosophy governing the thermal battery and multi-vessel load balancing. Referenced from Step 10. Intended for controls engineers and mechatronics integrators.

Appendix E — PLC Implementation and Discrete-Time Control Formulations — The mathematical implementation detail for the discrete-time automation control loops. Referenced from Appendix D. Intended for controls engineers writing PLC code.

Feedback Loops

The framework contains three formal iteration loops that return the designer to an earlier step when a constraint is violated. These are not failures of the design process — they are features of it. The loops are:

The Hydraulic Velocity Loop — Triggered in Step 5 when the velocity ceiling cannot be satisfied by VFD adjustment alone. Returns to Step 4 to increment the NPS.

The Thermal Deficiency Loop — Triggered in Steps 6 and 7 when the coil cannot deliver Q_{design} within the vessel envelope. Returns to Step 4 to increment the NPS.

The Q_{loss} Feedback Loop — Triggered in Step 9 when the rigorous heat loss calculation deviates from the placeholder by more than fifteen percent. Returns to Step 1 to recalculate Q_{design} with the updated loss value.

First-Time Users

Engineers using the VCH framework for the first time are advised to read the Framework Overview and the Glossary of Terms before beginning the core methodology. The three temperature differential definitions — ΔT_{coil} , ΔT_{drive} and ΔT_{loss} — are foundational to understanding how the thermal calculations relate to each other and should be understood clearly before Step 1 is started.

Experienced Users

For engineers familiar with the framework executing a subsequent design, the Step Output Summaries at the end of each step provide a compact checklist of all values that must be carried forward. The verification summary record in Step 11 provides the complete audit checklist for design sign-off.

Overview

Revised Methodology — First Edition

Preface

The Velocity-Constrained Hydronic (VCH) Sizing Framework is an iterative, constraint-driven engineering methodology for the thermal and hydraulic design of submerged hydronic heating systems operating within bulk liquid process vessels. The framework is validated for single-phase, closed-loop liquid-to-liquid heat exchange applications involving sensible heat transfer, with particular applicability to high-viscosity fluid thermal processing systems.

The methodology presented herein supersedes the linear design approach historically applied to such systems. Rather than treating thermal load calculation, pipe sizing, and pump selection as sequential independent tasks, the VCH framework integrates these disciplines into a unified, closed-loop optimization workflow governed by hard physical and operational boundaries. The design target is established independently from the hardware at the outset, ensuring that thermal adequacy is verified against an external process requirement rather than derived from the system's own capacity.

A note on cooling applications: the thermodynamic and hydraulic principles underlying this framework are applicable to cooling loops with symmetric logic. Where differences in constraint direction, fouling behaviour, or control safety logic apply to cooling configurations, these are identified explicitly within the relevant sections. The framework is oriented primarily toward industrial heating.

Governing Design Philosophy

The framework operates on four foundational principles that distinguish it from conventional sizing practice.

First, the thermal target is defined before the hardware. The required process heat load, multiplied by an appropriate factor of safety to account for long-term fouling, cold-start transients, and performance degradation, establishes a fixed design target against which all subsequent hardware decisions are evaluated. This ensures that the final system is verified against an independently defined requirement and not merely sized to its own theoretical maximum.

Second, all pipe geometry is locked to commercially available Nominal Pipe Sizes in accordance with ASME B36.10M and B36.19M. Arbitrary theoretical diameters are not permitted. This constraint ensures manufacturability, simplifies procurement, and anchors the hydraulic calculations to real physical dimensions from the outset.

Third, fluid velocity is treated as a hard ceiling, not a guideline. The maximum allowable velocity is determined by the selected thermofluid, the pipe material, and the erosion characteristics of the system. All pump selection, Variable Frequency Drive configuration, and control automation logic is bounded by this ceiling. The system is never permitted to operate above it under any steady-state condition.

Fourth, the framework applies a unified worst-case ambient reference temperature throughout. For heating applications, this is the historically coldest recorded ambient temperature at the installation site. For cooling applications, it is the historically hottest. This single reference point anchors the heat loss calculations, the thermal driver sizing, and the allowable thermofluid temperature differential within the circulation loop.

Temperature Differential Definitions

Three distinct temperature differentials govern the framework. Each serves a separate physical purpose and must not be conflated.

ΔT_{drive} is the temperature difference between the thermofluid inside the coil and the process fluid in the tank. This is the thermal driving force across the coil wall and governs the rate of heat transfer through the exchange surface. It is used in logarithmic mean temperature difference and Effectiveness-NTU calculations.

ΔT_{coil} is the temperature difference between the thermofluid at the coil inlet and the thermofluid at the coil outlet. This represents the energy deposited into the process vessel per unit mass of circulating fluid. To ensure that fluid thermophysical properties remain sufficiently uniform for single-pass lumped-parameter calculations to remain valid, ΔT_{coil} is capped at ten percent of the difference between the operating temperature and the worst-case ambient temperature for the installation site.

ΔT_{loss} is the temperature difference between the process fluid and the worst-case ambient temperature. This drives the steady-state environmental heat loss through the vessel walls and insulation and is used to size the thermal driver and quantify standing losses.

Framework Execution Sequence

Step 1 — Establish the Design Thermal Target

The design process begins with the client's process heat requirement. The total sensible energy required to bring the process fluid from its initial condition to its target operating temperature, compressed into the active operating window, defines the required continuous thermal power. This figure is then multiplied by the design factor of safety to establish the framework's fixed design target. This target is independent of all hardware decisions and serves as the verification benchmark throughout the design process. A minimum factor of safety of 1.5 is recommended for standard industrial applications, with 2.0 considered the practical optimum balancing operational resilience against material and capital cost.

Step 2 — Select and Characterise the Thermofluid

The secondary thermal transport medium is selected based on the operating temperature range, viscosity profile, material compatibility, thermal stability, and regulatory constraints of the application. Thermophysical properties — density, specific heat capacity, dynamic viscosity, and thermal conductivity — are extracted at the bulk operating temperature for use throughout the hydraulic and thermal calculations. For high-viscosity fluids, property variation with temperature should be noted, as this directly influences the conservatism required in heat transfer coefficient estimation.

Step 3 — Develop a Conceptual Vessel Geometry

A baseline vessel geometry is established from the client's target process volume. Standard compound vessel geometries — cylindrical shells with conical or hemi-ellipsoidal caps — are modelled using standard volumetric relationships. An aspect ratio appropriate to the application and structural requirements is selected, and the resulting internal diameter and shell height define the geometric canvas within which the thermal coil will be designed.

Step 4 — Select Nominal Pipe Size and Design the Maximum Coil Configuration

A commercially available pipe schedule is selected based on the anticipated hydraulic loading and fabrication requirements. Within the vessel envelope, a maximum helical coil configuration is modelled assuming a tightly wound geometry, subject to a minimum radial clearance from the vessel wall. The theoretical maximum coil length is then de-rated by a compound packing efficiency factor accounting for fabrication tolerances, structural support requirements, weld access clearance, thermal expansion allowance, and flow dead-zone prevention. The result is the maximum practical coil length physically achievable within the vessel.

Step 5 — Select a Pump for the Maximum Coil Configuration

Using the maximum practical coil geometry, the hydraulic resistance of the system is characterised across a representative range of operating flow rates using the Darcy-Weisbach framework with appropriate helical curvature corrections. A candidate pump is selected by overlaying its manufacturer performance curve onto the

system resistance curve to identify the natural operating point. The fluid velocity at this operating point is evaluated against the allowable velocity ceiling. If the ceiling is exceeded, the pipe diameter is incremented to the next standard NPS and the hydraulic analysis is repeated. Variable Frequency Drive scaling is applied where necessary to clamp the operating velocity below the threshold.

Step 6 — Determine Total Transmissible Thermal Power

At the stabilised operating flow rate, the maximum thermal power that the circulation loop is capable of delivering to the process vessel is evaluated. This accounts for the finite heat exchange capacity of the coil surface, the overall heat transfer coefficient under conservative worst-case viscosity and fouling conditions, and the practical limitations of the fluid's thermal transport capacity. The result represents the upper physical boundary of the system's thermal output.

Step 7 — Optimise the Coil and Pump to Match the Design Target

The maximum transmissible thermal power established in Step 6 is compared against the design thermal target established in Step 1. Where the hardware capacity significantly exceeds the target — as is typically the case when starting from a maximum coil configuration — the coil length is reduced iteratively until the transmissible thermal power converges to the design target. The coil geometry is back-calculated to determine the final manufacturable pitch and spacing. Following coil optimisation, the pump specification is reviewed and reduced where the lower hydraulic resistance of the shortened coil permits a smaller or less powerful unit.

Step 8 — Correct Vessel Volume for Coil Displacement

The physical volume displaced by the submerged coil assembly, support structures, and associated fittings is quantified. The vessel dimensions are then adjusted to recover this displaced volume, ensuring that the fabricated vessel delivers the full net process capacity specified by the client. The designer may expand the vessel radially or extend the cylindrical shell height depending on spatial constraints and manufacturing preference, with the latter preferred where floor footprint is limited.

Step 9 — Size the Thermal Driver

The total capacity required of the primary heating source — whether electric immersion elements, a fired thermal fluid heater, or another prime mover — is established as the sum of the optimized process heat load and the steady-state environmental heat loss across all vessels in the installation. For multi-vessel installations, the thermal driver must satisfy the cumulative demand across all active process nodes simultaneously. This sizing step yields both a maximum capacity figure, governing procurement, and a minimum capacity figure, governing the control system's lower operating boundary.

Step 10 — Size the Thermal Battery

The utility fluid reservoir that supplies heated thermofluid to the coil network is sized between a minimum and maximum volume boundary. The minimum is governed by the mechanical and control constraints of the selected thermal driver — ensuring adequate fluid mass to prevent element dry-firing, burner short-cycling, or compressor

anti-recycle violations — and by the priming volume of the closed hydronic loop. The maximum is governed by the peak batch thermal demand and the allowable temperature drawdown of the thermofluid. The designer selects a target volume within this range based on available capital, physical footprint constraints, and the desired degree of thermal stability and control responsiveness. A reservoir sized toward the upper boundary provides greater thermal inertia and smoother automation behaviour; one sized toward the lower boundary demands faster-acting control hardware to compensate for the reduced thermal buffer.

Step 11 — Verify Thermofluid Circulation Volume

With the thermal battery volume established, the allowable temperature differential of the thermofluid across the coil inlet and outlet is verified against the framework's ΔT_{coil} constraint. The temperature difference between the thermofluid entering and leaving the coil must be non-zero — confirming that energy transfer is occurring — and must not exceed ten percent of the difference between the operating temperature and the worst-case ambient temperature. This constraint ensures that fluid thermophysical properties remain within the range assumed during the hydraulic and thermal calculations. From the verified ΔT_{coil} , the required thermofluid mass in circulation and the corresponding reservoir volume contribution are confirmed.

Closed-Loop Iteration Logic

The framework incorporates three formal feedback paths that are invoked when a constraint is violated during the design sequence.

If the fluid velocity at the natural operating point exceeds the allowable ceiling and cannot be resolved by Variable Frequency Drive reduction without degrading pump head below safe operating limits, the designer returns to Step 4, increments the Nominal Pipe Size upward by one standard dimension, and repeats the hydraulic characterisation.

If the optimised transmissible thermal power remains insufficient to meet the design target despite coil maximisation, the primary bottleneck is volumetric throughput. The designer returns to Step 4, selects an expanded pipe diameter to permit higher flow rates at identical velocity thresholds, and re-evaluates the system.

If the corrected vessel volume required to offset coil displacement exceeds the available plant footprint, the designer returns to Step 8 and prioritises vertical shell extension over radial expansion to recover the required capacity within the spatial envelope.

This overview presents the conceptual and logical structure of the VCH framework. Governing equations, thermophysical property reference data, commercial pipe dimensional standards, and material selection matrices are compiled in the associated appendices.

VCH Method — Core Methodology

Step 1: Establishment of the Design Thermal Target

Overview

The design process begins not with hardware selection, but with a precise, independently defined thermal target. This distinction is fundamental to the VCH framework's integrity. All subsequent pipe sizing, coil geometry, pump selection, and thermal driver procurement decisions are evaluated against this target as a fixed external benchmark. It is never derived from the system's own hardware capacity.

Step 1 establishes two distinct thermal power demands that characterise the two primary operating modes of any batch thermal processing installation. The designer evaluates both and carries the governing case forward.

1.1 The Two Operating Modes

Industrial thermal processing systems do not operate under a single uniform load condition. Two physically distinct demand states govern the system at different points in the production cycle, and a robust design must account for both.

Batch Processing Mode represents the peak thermal demand condition. It occurs when raw material is introduced into the process vessel and must be brought from its incoming condition to the target operating temperature within a defined processing window. The energy requirement during this mode is substantial, as it must overcome both the sensible thermal deficit of the incoming material and, where applicable, the latent energy barrier of a phase transition.

Balance Mode represents the steady-state minimum demand condition. It occurs when the vessel is already at operating temperature and the system is simply maintaining thermal equilibrium against continuous environmental heat loss through the vessel boundaries. No material is being actively processed. The thermal load in this mode is comparatively low but must be sustained indefinitely for as long as the vessel remains active.

In most industrial installations, batch processing mode will govern the design. However, the framework evaluates both explicitly, as installations with extremely well-insulated vessels processing low-density materials at near-ambient temperatures may present exceptions.

1.2 Batch Target — Q_{batch}

The batch thermal target quantifies the continuous thermal power required to complete the processing of a defined incoming material quantity within the specified operational window.

1.2.1 Incoming Material Condition

The framework assumes a conservative incoming material condition: the raw material arrives at the worst-case ambient temperature for the installation site. For heating applications, this is the historically coldest recorded ambient temperature. This assumption eliminates the risk of undersizing the system for cold delivery conditions, which are common in unheated storage yards, open logistics environments, and seasonal climates.

1.2.2 Total Batch Energy Components

The total energy required to process one batch, Q_{total} , is the sum of all thermal contributions required to transition the material from its incoming condition to its target operating state.

For materials that undergo a solid-to-liquid phase transition within the operating range — a common characteristic of heavy organic commodities, waxes, polymer feedstocks, and certain edible fats — the total batch energy comprises three sequential contributions:

Component 1 — Sensible Heating of the Solid Phase

The energy required to raise the incoming solid material from the worst-case ambient temperature to its melting point:

$$Q_{solid} = m \times C_{p_{solid}} \times (T_{melt} - T_{worst\ ambient})$$

Where:

- m is the total batch mass (kg), as specified by the designer
- $C_{p_{solid}}$ is the specific heat capacity of the material in its solid phase (J/kg·K)
- T_{melt} is the onset melting temperature of the material (°C)
- $T_{worst\ ambient}$ is the worst-case incoming material temperature (°C)

Component 2 — Latent Heat of Fusion

The energy required to complete the phase transition from solid to liquid at the melting point. This contribution carries no associated temperature rise; it represents the energy absorbed by the material's molecular structure during the transition:

$$Q_{melt} = m \times L_f$$

Where:

- L_f is the latent heat of fusion of the material (J/kg)

For materials that do not undergo a phase transition within the operating range, this component is zero and may be omitted.

Component 3 — Sensible Heating of the Liquid Phase

The energy required to raise the fully melted material from the melting point to the target process operating temperature:

$$Q_{liquid} = m \times C_{p_{liquid}} \times (T_{operating} - T_{melt})$$

Where:

- $C_{p_{liquid}}$ is the specific heat capacity of the material in its liquid phase (J/kg·K)
- $T_{operating}$ is the target process fluid temperature (°C)

For materials with no phase transition, Components 1 and 3 combine into a single sensible heat term evaluated across the full temperature range from ambient to operating temperature, and T_{melt} is replaced by $T_{worst\ ambient}$ as the lower bound of Component 3.

Total Batch Energy:

$$Q_{total} = Q_{solid} + Q_{melt} + Q_{liquid}$$

Example: A facility processing palm oil — which melts at approximately 35°C and arrives solid during winter months — would evaluate all three components. A facility processing a liquid petroleum derivative that arrives above its pour point year-round would evaluate Components 1 and 3 only, with T_{melt} replaced by $T_{worst\ ambient}$.

1.2.3 Conversion to Continuous Thermal Power

The total batch energy is distributed across the available processing window to yield a continuous thermal power requirement. The default operational window is seven hours, representing a standard single industrial shift. Where the designer specifies a shorter processing window — for example, to meet higher throughput targets — t_{run} is reduced accordingly, and the required continuous power increases proportionally.

$$Q_{batch} = \frac{Q_{total}}{t_{run}}$$

Where:

- t_{run} is the available processing duration (seconds); default 7 hours = 25,200 seconds

The designer should note that specifying an aggressively short processing window has a compounding effect: it increases Q_{batch} , which increases Q_{design} , which drives larger coil geometry, higher pump capacity, and a more powerful thermal driver. The t_{run} specification is therefore a capital cost lever, not merely a scheduling parameter.

1.3 Balance Target — $Q_{balance}$

The balance thermal target quantifies the continuous thermal power required to maintain a vessel already at its target operating temperature against steady-state environmental heat loss. This represents the minimum sustained output the thermal driver must be capable of delivering during idle holding periods between active batch cycles.

At this stage of the design process, the vessel geometry and insulation specification have not yet been finalised. A precise thermal resistance network calculation is therefore not yet possible. The framework applies a calibrated placeholder estimate based on the insulation tier selected by the designer. This estimate is replaced by a rigorous thermal resistance calculation at the thermal driver sizing stage.

The environmental heat loss rate is estimated as a fraction of the total thermal energy stored within the process fluid at operating temperature:

$$Q_{balance} = k \times m_{product} \times C_{P_{liquid}} \times T_{operating}$$

Where:

- $m_{product}$ is the total mass of process fluid held in the vessel at operating temperature (kg)
- $C_{P_{liquid}}$ is the specific heat capacity of the process fluid in its liquid phase (J/kg·K)
- $T_{operating}$ is the target process fluid temperature (°C)
- k is the fractional thermal loss rate per second, determined by the insulation tier
-

Insulation Tier	Description	k (per hour)	k (per second)
Well Insulated	Modern mineral wool or polyurethane foam cladding, fully sealed and weather-protected.	2%	5.56×10^{-6}
Lightly Insulated	Basic fibreglass wrap or partial cladding.	5%	1.39×10^{-5}
Uninsulated	Bare metal vessel with no thermal cladding. Represents the worst-case heat loss scenario.	12%	3.33×10^{-5}

Upper bound values are applied throughout. If the actual insulation specification is not yet determined, the

designer should select the Lightly Insulated tier as a conservative default.

The designer should note that $Q_{balance}$ will be revisited and replaced with a rigorously calculated value during the thermal driver sizing phase. If the rigorously calculated Q_{loss} at that stage yields a value that renders the system thermally inadequate — that is, if the actual heat loss substantially exceeds the placeholder estimate — the design must be returned to this step and the thermal target recalculated before proceeding.

1.4 Governing Thermal Power and Design Target

With both operating mode demands established, the governing thermal power is taken as the larger of the two:

$$Q_{required} = \max(Q_{batch}, Q_{balance})$$

In the overwhelming majority of industrial batch processing installations, Q_{batch} will govern. $Q_{balance}$ governs only in installations where the vessel is exceptionally large, the operating temperature is very high, the insulation is poor, and the batch throughput is comparatively modest.

The design thermal target — the fixed independent benchmark against which all hardware in this framework is evaluated — is then established by applying the design factor of safety:

$$Q_{design} = FOS \times Q_{required}$$

The factor of safety accounts for long-term thermal performance degradation due to surface fouling, uncertainty in heat transfer coefficient estimation, manufacturing tolerance effects on coil geometry, and the elevated thermal demand of cold-start conditions. The following bounds apply:

FOS Value	Application
1.5	Minimum permissible. Applicable only where fluid fouling potential is low and operating conditions are highly stable.
2	Recommended standard. Appropriate for the majority of industrial heating applications involving organic process fluids.
2.5	Maximum recommended.

Q_{design} is carried forward unchanged through all subsequent steps of the framework. It is not adjusted, re-derived, or overridden by hardware capacity findings at any later stage.

1.5 Step 1 Output Summary

Upon completion of Step 1, the designer has established the following:

- Q_{batch} : the continuous thermal power required to process the specified batch within the defined

operational window, accounting for all phase and sensible energy contributions from worst-case ambient conditions

- $Q_{balance}$: the estimated continuous thermal power required to maintain the vessel at operating temperature against environmental loss during idle holding periods
- $Q_{required}$: the governing thermal power demand
- Q_{design} : the fixed design thermal target, inclusive of the factor of safety, against which all subsequent hardware decisions are verified

These four values, together with the worst-case ambient temperature and the insulation tier selection, constitute the complete output of Step 1 and serve as the primary inputs to the thermofluid selection and vessel conceptualisation steps that follow.

Step 2: Thermofluid Selection and Property Characterisation

Overview

The selection of the primary thermal transport medium is one of the most consequential decisions in the design process. The thermofluid is the working substance that carries thermal energy from the heating source through the internal coil network and into the process vessel. Every subsequent calculation in the framework — hydraulic resistance, pump sizing, heat transfer coefficient estimation, coil surface area, and thermal battery volume — is directly conditioned by the thermophysical properties of this fluid.

A poor thermofluid selection cannot be corrected by oversizing the coil or upgrading the pump. It propagates as a fundamental constraint through the entire design. The designer must therefore approach this selection with the same rigour applied to the thermal target established in Step 1.

Step 2 produces a fully characterised thermofluid specification: the selected fluid, its operating temperature range, and its thermophysical properties evaluated at the bulk operating condition. These properties are carried forward as fixed inputs to all subsequent steps.

2.1 Selection Criteria

The thermofluid must satisfy the operational, chemical, regulatory, and economic constraints of the specific installation. The following criteria govern the selection decision and must each be evaluated before a fluid is confirmed.

Operating Temperature Range

The thermofluid must remain in a stable single liquid phase throughout its entire operating cycle — from cold standby conditions through to peak operating temperature. The fluid's lower operating limit must comfortably exceed its freezing or pour point under the worst-case ambient conditions of the installation site. Its upper operating limit must remain below its boiling point at system operating pressure, and below the fluid's rated thermal degradation threshold with an adequate margin. Proximity to either boundary introduces phase instability, vapour lock risk, and accelerated fluid degradation.

Viscosity Profile

Dynamic viscosity governs the hydraulic resistance of the circulation loop, the Reynolds number regime within the coil, and the internal convective heat transfer coefficient. Fluids with high viscosity at operating temperature produce elevated pressure drops, unfavourable laminar flow conditions, and reduced heat transfer performance. The viscosity profile across the full operating temperature range must be reviewed, not merely the value at peak temperature. For systems that undergo cold starts from ambient conditions, the viscosity at the lowest expected startup temperature determines the worst-case hydraulic loading on the pump.

Thermal Stability

Thermofluids subjected to prolonged high-temperature operation are susceptible to oxidative degradation, thermal cracking, and polymerisation. These degradation mechanisms increase fluid viscosity over time, reduce thermal conductivity, deposit insulating fouling layers on internal pipe surfaces, and in severe cases generate corrosive breakdown products. The designer must confirm that the selected fluid's rated continuous operating temperature comfortably exceeds the system's peak operating requirement, and must establish a fluid monitoring and replacement schedule appropriate to the application.

Material Compatibility

The thermofluid must be chemically compatible with all wetted materials in the circulation loop — including the internal coil pipe alloy, pump casing and seal materials, expansion vessel lining, and any inline instrumentation wetted surfaces. Incompatibility manifests as accelerated corrosion, seal degradation, or metallic contamination of the fluid, each of which compromises system integrity and fluid thermal performance over time. The pipe material confirmed in Step 4 must be cross-checked against the thermofluid selection at that stage.

Regulatory and Safety Classification

In food processing, pharmaceutical, and potable water adjacent applications, the thermofluid must carry an appropriate food-grade or regulatory certification. In enclosed or poorly ventilated plant environments, flash point and vapour toxicity ratings must be evaluated against local safety codes. The designer must confirm that the selected fluid's safety classification is consistent with the installation environment before proceeding.

Economic Considerations

Fluid cost encompasses not only initial procurement but also the volume required to prime the full circulation loop, ongoing top-up requirements due to evaporative or leakage losses, disposal costs at end of service life, and the cost implications of the fluid monitoring programme. Higher-performance specialty fluids frequently offer a favourable lifecycle cost relative to cheaper alternatives that degrade rapidly or impose higher maintenance burdens.

2.2 Common Thermofluid Classifications

The following classifications represent the principal thermofluid families encountered in industrial hydronic heating applications. Detailed thermophysical property tables for each fluid class are compiled in Appendix B and must be consulted during the property characterisation step that follows.

Industrial Treated Water

Applicable operating range of 0°C to 100°C at atmospheric pressure, extendable under pressurised system conditions. Water offers the highest specific heat capacity of any common thermofluid, low viscosity across its operating range, and negligible cost. It is the preferred selection for low-to-moderate temperature applications where freeze protection is not required. Corrosion inhibitor treatment is mandatory in closed steel loop systems to prevent internal oxidation scaling.

Ethylene Glycol — Water Solutions

Applicable operating range of approximately –20°C to 120°C depending on mixture concentration. The addition of ethylene glycol depresses the freeze point and elevates the boiling point relative to pure water, at the cost of reduced specific heat capacity and increased viscosity. The 50% volumetric mixture represents the most commonly specified configuration, balancing freeze protection against hydraulic performance. This fluid class is the standard selection for outdoor or unheated plant installations in temperate and cold climates.

High-Temperature Synthetic Thermal Oils

Applicable operating range of approximately 50°C to 300°C or above depending on fluid grade. Synthetic thermal oils are the standard selection for process temperatures exceeding the safe pressurised water ceiling. They offer excellent thermal stability at elevated temperatures and low vapour pressure, eliminating the pressurisation requirements of high-temperature water systems. Their primary disadvantages are significantly higher viscosity at low temperatures — which imposes severe hydraulic loading during cold starts — lower specific heat capacity relative to water-based fluids, and substantially higher procurement and disposal cost. Therminol 66 and Dowtherm A represent widely used industrial benchmarks within this classification.

Food-Grade Thermal Oils

Applicable operating range consistent with synthetic thermal oils. Food-grade thermal oils carry regulatory certification confirming that incidental product contact does not present a food safety hazard. They are the mandatory selection for direct food contact or food-adjacent processing environments, including edible oil

refining, confectionery manufacturing, and dairy processing. Performance characteristics are broadly comparable to synthetic thermal oils of equivalent viscosity grade, at a higher procurement cost reflecting the certification overhead.

Example: A palm oil melting and storage installation operating at 60°C to 80°C with food-grade regulatory requirements would typically select a certified food-grade thermal oil. An industrial chemical processing vessel operating at 200°C with no food contact risk would select a high-temperature synthetic thermal oil on economic grounds.

2.3 Thermophysical Property Characterisation

Once the thermofluid is selected, its thermophysical properties must be extracted at the bulk operating temperature of the circulation loop. These properties are used directly in the hydraulic resistance calculations, the heat transfer coefficient estimation, and the thermal battery sizing steps that follow.

The four primary thermophysical properties required by the framework are:

Density (ρ) — governs the mass flow rate corresponding to a given volumetric flow rate, and contributes to hydraulic pressure calculations. Units: kg/m³.

Specific Heat Capacity (C_p) — governs the thermal energy carried per unit mass per degree of temperature change. Directly determines the mass flow rate required to deliver Q_{design} . Units: J/(kg·K).

Dynamic Viscosity (μ) — governs the Reynolds number, the flow regime classification, the hydraulic friction factor, and the internal convective heat transfer coefficient. Units: mPa·s.

Thermal Conductivity (k) — governs the conductive component of the overall heat transfer coefficient through the pipe wall and fluid boundary layers. Units: W/(m·K).

These properties are extracted from the reference tables in Appendix B at the bulk mean operating temperature of the thermofluid within the circulation loop. The bulk mean temperature is defined as:

$$T_{bult} = \left(\frac{T_{supply} - T_{return}}{2} \right)$$

Where T_{supply} is the thermofluid temperature entering the coil and T_{return} is the thermofluid temperature leaving the coil. At this stage of the design process, T_{return} is not yet precisely known. The designer should use $T_{operating}$ as an initial approximation for T_{supply} and apply the ΔT_{coil} constraint established in the framework overview to bound T_{return} . The bulk mean temperature is then estimated accordingly and refined in later steps as the design converges.

2.3.1 Interpolation Between Tabulated Values

Appendix B provides thermophysical properties at discrete temperature intervals. Where the bulk mean operating temperature falls between two tabulated reference points, the property value is determined by linear interpolation:

$$P(T) = P_1 + \left[\frac{(T-T_1)}{(T_2-T_1)} \right] \times (P_2 - P_1)$$

Where:

- $P(T)$ is the property value at the required temperature T
- P_1 and P_2 are the tabulated property values at the bounding temperatures T_1 and T_2 respectively

This interpolation is applied independently to each of the four thermophysical properties. The full mathematical basis for temperature-dependent property variation and the limits of linear interpolation accuracy across wide temperature intervals are discussed in Appendix A.

2.3.2 Conservative Viscosity Selection

Dynamic viscosity warrants particular attention. For the purposes of heat transfer coefficient estimation, the framework requires that viscosity be evaluated not only at the bulk mean operating temperature but also at the worst-case condition — specifically the lowest temperature the thermofluid will experience during normal operation, which corresponds to the cold-start condition at system startup.

The worst-case viscosity value is used in the overall heat transfer coefficient calculation to ensure that the coil surface area is sized for the most thermally demanding operating condition the system will encounter. The bulk mean viscosity is used in the hydraulic resistance and Reynolds number calculations under steady-state operating conditions.

This distinction is critical for high-viscosity thermofluid classes — particularly synthetic and food-grade thermal oils — where viscosity at cold-start temperatures may be an order of magnitude higher than at the nominal operating temperature. Failing to account for this in the heat transfer sizing will result in a coil that is adequate at steady state but thermally insufficient during startup.

2.4 Downstream Constraints Flagged by Thermofluid Selection

The thermofluid selected in this step introduces constraints that are resolved at later stages of the framework. The designer should note the following dependencies before proceeding.

Velocity Ceiling (v_{max}) — The maximum allowable fluid velocity within the internal coil is determined by the combination of thermofluid properties and pipe material. This ceiling is evaluated and confirmed in Step 4 once the pipe schedule and material are selected. The thermofluid choice influences this ceiling through its density,

viscosity, and chemical compatibility with candidate pipe materials. No numerical velocity limit is assigned at this stage.

Pipe Material Compatibility — The chemical compatibility of the selected thermofluid with candidate pipe materials must be formally verified in Step 4. The designer should carry a preliminary note of any known incompatibilities identified during the selection process in Section 2.1 into that step. Material compatibility data is compiled in Appendix C.

Pump Cavitation Risk — The thermofluid's vapour pressure at its maximum operating temperature determines the available Net Positive Suction Head margin of the circulation pump. Fluids operating close to their vapour pressure ceiling impose stricter constraints on pump inlet conditions and system pressurisation requirements. This is evaluated during pump selection in Step 5.

2.5 Step 2 Output Summary

Upon completion of Step 2, the designer has established the following:

- The selected thermofluid and its regulatory classification
- The confirmed operating temperature range of the circulation loop
- The four thermophysical properties — ρ , C_p , μ , k — evaluated at the bulk mean operating temperature
- The worst-case viscosity value for conservative heat transfer coefficient estimation
- A preliminary note of any pipe material compatibility constraints and cavitation risk factors to be resolved in Steps 4 and 5

These outputs are carried forward as fixed fluid inputs to all subsequent hydraulic, thermal, and mechanical sizing steps.

Step 3: Conceptual Vessel Geometry and Baseline Dimensional Framework

Overview

Step 3 establishes the physical spatial envelope of the primary process vessel. Before a thermal coil can be designed, the geometric boundaries of the process fluid must be mathematically defined. The vessel dimensions derived in this step serve a dual purpose: they provide the geometric canvas within which the internal coil assembly will be packaged in subsequent steps, and they define the external surface area through which environmental heat loss occurs — directly linking back to the Q_{loss} placeholder established in Step 1.

This step produces a fully defined baseline vessel geometry characterised by its internal diameter, shell height, cap profile, total volume, and external surface area. These dimensions are treated as the starting point for the design. They will be revisited and adjusted in a later step to compensate for the volumetric displacement introduced by the submerged coil assembly.

3.1 Vessel Geometry Selection

Industrial process vessels for bulk liquid thermal management are standardised as compound geometric solids. The most common configuration consists of a primary cylindrical shell body capped at the top and bottom with identical geometric profiles. This compound form balances structural efficiency, fabrication simplicity, and thermal performance in a way that bespoke geometries cannot match at comparable cost.

The designer must make two primary geometric decisions at this stage: the selection of the cap profile and the establishment of the vessel aspect ratio.

3.1.1 Cap Profile Selection

Two cap profiles are supported within the VCH framework. The selection criterion is governed by the operating pressure requirement and the drainability needs of the process.

Conical Caps

Conical caps are the standard selection for atmospheric or near-atmospheric pressure vessels where complete gravity drainage of the vessel contents is operationally required. The converging geometry of the cone directs residual fluid toward the centreline drain point, achieving substantially complete drainage without mechanical assistance. This is particularly advantageous in batch processing applications where vessel cleaning between product runs is required, or where complete product recovery is economically significant.

Conical caps are simpler and less expensive to fabricate than hemi-ellipsoidal profiles, as they require only flat plate rolling and welding rather than compound curvature forming. However, the stress concentration at the

cone-to-cylinder junction requires careful weld detailing, and conical caps are structurally unsuited to elevated internal pressures without reinforcement rings.

Hemi-Ellipsoidal Caps

Hemi-ellipsoidal caps are the standard selection for pressurised process vessels. The ellipsoidal profile distributes internal pressure stresses uniformly across the cap surface, eliminating the localised stress concentrations inherent in conical geometry. This makes hemi-ellipsoidal caps the preferred choice for closed-loop systems operating above atmospheric pressure, high-temperature applications where internal vapour pressure is significant, and any installation governed by pressure vessel design codes.

The fabrication cost of hemi-ellipsoidal caps is higher than conical equivalents due to the compound curvature forming requirements. For atmospheric applications where drainage completeness is not a primary concern, the hemi-ellipsoidal profile remains structurally acceptable but is rarely specified on economic grounds.

The designer selects one cap profile and applies it symmetrically to both the top and bottom of the vessel. Mixed cap configurations are not addressed within this framework.

3.2 Aspect Ratio and Dimensional Relationships

The vessel aspect ratio defines the relationship between the primary cylindrical shell height and the cap height. This ratio has structural, thermal, and hydraulic implications that extend beyond simple geometry.

A vessel that is excessively wide relative to its height presents a large plan footprint, produces unfavourable thermal stratification patterns in high-viscosity fluids due to the shallow fluid depth, and limits the number of helical coil turns achievable within the cylindrical section. A vessel that is excessively tall relative to its diameter increases structural shell loading, complicates internal coil support arrangements, and elevates fabrication cost through increased weld length and more complex site erection requirements.

The framework establishes a standard aspect ratio between the cylindrical shell height h_1 and the cap height h_2 :

$$\frac{h_2}{h_1} = \frac{1}{4} \rightarrow h_1 = 4h_2$$

This ratio reflects established industrial practice for heated bulk storage vessels and produces a geometry that balances coil packing capacity, structural efficiency, thermal circulation behaviour, and fabrication economy. The total vessel height h is therefore:

$$h = 2h_1 + h_2 \Rightarrow 2(4h_2) + h_2 \Rightarrow 9h_2$$

3.2.1 Aspect Ratio Sanity Check

Before proceeding to volume calculation, the designer should verify that the resulting vessel geometry is practically realisable within the constraints of the installation site and the fabrication shop.

The following boundary conditions define the practical geometric limits of the framework:

Minimum diameter: The vessel internal diameter d must be sufficient to accommodate the internal coil assembly with the required minimum radial clearance on all sides. A practical lower bound of 600mm internal diameter is recommended. Vessels smaller than this impose severe restrictions on coil geometry and internal maintenance access.

Maximum height-to-diameter ratio: A height-to-diameter ratio (h/d) exceeding 4.0 indicates a vessel that is disproportionately tall relative to its footprint. Such geometries impose elevated structural loads on the shell and foundation, complicate coil installation and internal inspection, and are atypical of standard industrial fabrication practice. Where the calculated geometry exceeds this threshold, the designer should reconsider the aspect ratio or consult a structural engineer before proceeding.

Minimum height-to-diameter ratio: A height-to-diameter ratio below 0.8 indicates a vessel that is disproportionately wide and shallow. Such geometries provide limited cylindrical shell height for coil winding, are susceptible to thermal stratification in viscous fluids, and may produce impractical coil configurations in subsequent steps. Where the calculated geometry falls below this threshold, the aspect ratio should be revised upward.

If the geometry falls outside these practical bounds after applying the standard aspect ratio, the designer may adjust h_2/h_1 within a reasonable range to recover a practicable geometry, provided the modification is documented and its implications for coil packing capacity are noted.

3.3 Baseline Volume Calculations

With the cap profile selected and the aspect ratio established, the baseline internal vessel volume is calculated from the client's target process volume V_{target} . The volume formulation differs between the two supported cap profiles.

3.3.1 Conical-Capped Cylindrical Vessel

For a cylindrical vessel with symmetrical conical top and bottom caps, the total internal volume is:

$$V_{tank} = \left(\frac{\pi d^2}{4} \right) \times \left(h_1 + \frac{2h_2}{3} \right)$$

Where:

- d is the internal vessel diameter (m)
- h_1 is the cylindrical shell height (m)
- h_2 is the individual conical cap height (m)
- The $h_3/3$ term represents the combined volumetric contribution of both conical caps

3.3.2 Hemi-Ellipsoidal-Capped Cylindrical Vessel

For a cylindrical vessel with symmetrical hemi-ellipsoidal top and bottom caps, the total internal volume is:

$$V_{tank} = \left(\frac{\pi d^2}{4} \right) \times \left(h_1 + \frac{2h_2}{3} \right)$$

Where the combined volumetric contribution of both hemi-ellipsoidal caps is given by:

$$V_{caps} = \left(\frac{\pi d^2 \times h_2}{6} \right)$$

Yielding the complete expression:

$$V_{tank} = \left(\frac{\pi d^2}{4} \right) \times h_1 + \left(\frac{\pi d^2 \times h_2}{6} \right)$$

3.3.3 Solving for Vessel Dimensions

Setting V_{tank} equal to the client's target process volume V_{target} and substituting the aspect ratio relationship $h_1 = 4h_2$, the designer solves for h_2 and subsequently derives d , h_1 , and h . This system of equations is solved analytically. The full algebraic derivation for each cap geometry is provided in Appendix A.

The resulting dimensions define the baseline vessel geometry. These are the dimensions before coil displacement compensation and represent the minimum internal volume envelope required to satisfy the client's process capacity requirement.

3.4 External Surface Area Calculation

The external surface area of the vessel is required to quantify environmental heat loss in the thermal driver sizing step and to verify the Q_{loss} placeholder estimate carried forward from Step 1. The total external surface area A_{total} is the sum of the contributions from the cylindrical shell and both caps.

3.4.1 Cylindrical Shell Surface Area

$$A_{shell} = \pi d h_1$$

3.4.2 Conical Cap Surface Area

For a single conical cap of base diameter d and height h_2 , the lateral surface area is:

$$A_{cone} = \frac{d\pi}{2} \times \sqrt{\left(\left(\frac{d}{2}\right)^2 + h_2^2\right)}$$

For two symmetrical conical caps:

$$A_{caps\ conical} = d\pi \times \sqrt{\left(\left(\frac{d}{2}\right)^2 + h_2^2\right)}$$

3.4.3 Hemi-Ellipsoidal Cap Surface Area

For a single hemi-ellipsoidal cap of base diameter d and height h_2 , the lateral surface area is approximated by:

$$A_{ellipse} = 2\pi \times \left(\frac{d}{2}\right)^2 \times \frac{\left[1 + \frac{(1-e^2) \times \text{arctanh}(e)}{e}\right]}{2}$$

Where the eccentricity e is defined as:

$$e = \sqrt{1 - \left(\frac{h_2}{d/2}\right)^2}$$

For two symmetrical hemi-ellipsoidal caps the result is doubled. The full derivation and a simplified working approximation suitable for preliminary design are provided in Appendix A.

3.4.4 Total External Surface Area

$$A_{total} = A_{shell} + A_{caps}$$

This value is carried forward to the thermal driver sizing step where it is used in the thermal resistance network to compute the actual steady-state heat loss rate Q_{loss} . If the resulting Q_{loss} deviates substantially from the placeholder estimate made in Step 1, the designer must return to Step 1 and recalculate $Q_{balance}$ and Q_{design} before proceeding further. This constitutes the formal closure of the Q_{loss} feedback loop introduced in Step 1.

3.5 Aspect Ratio Flexibility and Engineering Judgement

The standard aspect ratio $\frac{h_2}{h_1} = \frac{1}{4}$ is the recommended default and will produce a practical geometry for the majority of industrial applications. However, the designer retains the latitude to deviate from this ratio where site-specific constraints — such as a restricted ceiling height, a constrained floor footprint, or the need to match an existing vessel family — make the standard ratio impractical.

Any deviation from the standard aspect ratio must be documented with a clear engineering rationale, and its downstream implications must be assessed:

A reduction in h_1 relative to d reduces the available cylindrical shell height for coil winding and will directly constrain the maximum achievable coil length in Step 4. If the reduction is significant, the designer should perform a preliminary check at this stage to confirm that adequate coil length remains achievable before committing to the modified geometry.

An increase in h_1 relative to d increases the available coil winding height but may introduce thermal stratification concerns in high-viscosity process fluids and will increase structural shell loading. In tall, narrow vessels processing viscous fluids without mechanical agitation, the assumption of uniform bulk temperature made elsewhere in this framework becomes less conservative and should be noted.

3.6 Step 3 Output Summary

Upon completion of Step 3, the designer has established the following:

- The selected cap profile and the engineering rationale for the selection
- The baseline vessel internal diameter d , cylindrical shell height h_1 , cap height h_2 , and total height h
- The baseline internal vessel volume V_{tank} , confirming satisfaction of the client's target process volume V_{target}
- The external surface area A_{total} , comprising shell and cap contributions, for use in the thermal driver sizing step
- Confirmation that the vessel geometry satisfies the practical aspect ratio bounds defined in Section 3.2.1

These outputs define the geometric canvas for the coil design and hydraulic sizing steps that follow. They are subject to revision in the volumetric displacement compensation step once the internal coil geometry is finalised.

Step 4: Nominal Pipe Size Selection and Maximum Coil Configuration

Overview

Step 4 transitions the design from fluid and vessel characterisation into the first piece of physical hardware definition. Two things are accomplished here that have been deliberately deferred until this point. The maximum allowable fluid velocity is formally assigned as a hard numerical ceiling for the first time, having been flagged as a downstream constraint in Step 2. And the internal thermal coil is sized to its maximum physically achievable configuration within the vessel envelope established in Step 3.

The maximum coil configuration computed in this step represents the absolute upper boundary of the system's heat exchange capacity. It is not the final coil specification — that is determined in Step 7 through optimisation against Q_{design} . What Step 4 establishes is the physical ceiling within which that optimization operates. If the maximum coil configuration cannot deliver Q_{design} even at its upper limit, the design cannot proceed and the pipe diameter must be incremented before continuing.

4.1 Pipe Material Confirmation and Velocity Ceiling Assignment

The pipe material for the internal heat exchange coil must be confirmed before any geometric calculations are performed. The velocity ceiling v_{max} is a function of both the thermofluid selected in Step 2 and the pipe material selected here. These two decisions together determine the erosion threshold that governs the entire hydraulic design.

4.1.1 Material Selection Criteria

The pipe material must satisfy the chemical compatibility requirement established in Step 2 and must meet the thermal, mechanical, and hygienic demands of the application. The material evaluation matrix in Appendix C provides a comprehensive assessment of candidate materials across the following engineering criteria:

- Thermal conductivity — higher conductivity reduces the conductive resistance term in the overall heat transfer coefficient, directly improving coil thermal performance
- Corrosion and oxidation resistance — critical for long service life in chemically active or high-temperature environments
- Surface finish — smooth internal surfaces minimise hydraulic friction losses and resist fouling accumulation; smooth external surfaces maximise convective heat transfer to the process fluid
- Fabrication characteristics — weldability, bendability, and availability in helical coil configurations
- Regulatory compliance — food-grade certification where required by the application
- Mechanical durability under cyclic thermal loading

For the majority of industrial hydronic heating applications involving organic process fluids, austenitic stainless steel to ASTM A312 TP316L is the recommended baseline material. It offers a balance of corrosion resistance,

hygienic surface finish, fabrication practicality, and adequate thermal conductivity that is difficult to match at comparable lifecycle cost. Where thermal conductivity is the primary design constraint and fluid velocity can be strictly controlled, copper to ASTM B88 Type L offers substantially higher conductivity at the cost of a significantly more restrictive velocity ceiling.

4.1.2 Velocity Ceiling Assignment

With the pipe material confirmed, the maximum allowable fluid velocity v_{max} is formally assigned. This ceiling is determined by the erosion characteristics of the pipe material under the hydrodynamic conditions imposed by the selected thermofluid. It represents the absolute upper boundary for fluid velocity within the coil under all steady-state operating conditions. The circulation pump, Variable Frequency Drive configuration, and all automation control logic established in subsequent steps are bounded by this ceiling without exception.

The following velocity ceilings apply within this framework:

Pipe Material	Thermofluid Class	v_{max}
Stainless Steel (TP316L)	Water / Glycol Solutions	3.0 m/s
Stainless Steel (TP316L)	Thermal Oils	2.0 m/s
Carbon Steel (A106 Gr.B)	Water / Glycol Solutions	2.5 m/s
Carbon Steel (A106 Gr.B)	Thermal Oils	1.8 m/s
Copper (B88 Type L)	All thermofluid classes	1.5 m/s

These ceilings reflect established industry erosion thresholds for the stated material-fluid pairings under continuous hydronic service in helical coil configurations. The lower ceiling assigned to copper across all fluid classes reflects the heightened susceptibility of soft metal alloys to velocity-induced wall thinning along tight helical bends, where centrifugal forces concentrate fluid shear stress against the outer pipe wall. Detailed material property data and roughness coefficients are compiled in Appendix C.

The assigned v_{max} value is recorded and carried forward unchanged through all subsequent hydraulic and automation design steps.

4.2 Nominal Pipe Size Selection

With v_{max} established, the minimum internal flow area required to convey the target volumetric flow rate without breaching the velocity ceiling is determined. This provides an objective, deterministic basis for Nominal Pipe Size selection.

4.2.1 Target Volumetric Flow Rate

The target volumetric flow rate Q_{flow} is the minimum continuous circulation rate required to deliver Q_{design} to the process vessel within the operational window. It is derived from the design thermal target established in Step 1 and the thermofluid properties characterised in Step 2:

$$Q_{flow} = \frac{Q_{design}}{\rho \times C_p \times \Delta T_{coil}}$$

Where:

- Q_{design} is the fixed design thermal target from Step 1 (W)
- ρ is the thermofluid density at bulk operating temperature (kg/m^3)
- C_p is the thermofluid specific heat capacity at bulk operating temperature ($\text{J/kg}\cdot\text{K}$)
- ΔT_{coil} is the allowable thermofluid temperature differential across the coil inlet and outlet, capped at 10% of $(T_{operating} - T_{ambient\ worst})$ as established in the framework overview (k)

This flow rate represents the hydraulic throughput the system must achieve under steady-state operating conditions to satisfy the thermal design target.

4.2.2 Minimum Required Internal Flow Area

Enforcing the velocity ceiling v_{max} against the target volumetric flow rate yields the minimum internal cross-sectional area the pipe must provide:

$$A_{min} = \frac{Q_{flow}}{v_{max}}$$

Where:

- A_{min} is the minimum required internal flow area (m^2)
- Q_{flow} is the target volumetric flow rate (m^3/s)
- v_{max} is the velocity ceiling assigned in Section 4.1.2 (m/s)

4.2.3 Standard NPS Selection

The designer cross-references A_{min} against the internal flow area column of the ASME piping dimensional dataset in Appendix C. The selected Nominal Pipe Size is the smallest standard schedule that provides an internal flow area equal to or greater than A_{min} , ensuring that the actual fluid velocity v_{actual} at the target flow rate does not breach the velocity ceiling.

$$v_{actual} = \frac{Q_{flow}}{A_{selected}} \leq v_{max}$$

Where $A_{selected}$ is the internal flow area of the chosen NPS schedule (m^2).

The preference is always for the smallest NPS that satisfies the constraint. Unnecessarily large pipe diameters reduce fluid velocity, which reduces the Reynolds number and degrades the internal convective heat transfer coefficient — increasing the coil surface area required to achieve Q_{design} . Pipe sizing is therefore a balance between velocity constraint compliance and heat transfer performance, and the smallest compliant NPS consistently provides the best thermal outcome.

The selected NPS, its external diameter d_{pipe} , internal diameter d_{in} , wall thickness t_w , and internal flow area $A_{selected}$ are recorded and carried forward to all subsequent geometric and hydraulic calculations.

4.3 Maximum Coil Geometry

With the pipe geometry confirmed, the maximum physically achievable coil configuration within the vessel envelope is modelled. This establishes the upper boundary of the system's heat exchange surface area.

4.3.1 Maximum Coil Diameter

The internal coil must maintain a minimum radial clearance of 100mm between its outer edge and the internal vessel wall on all sides. This clearance is mandatory to prevent localised thermal stagnation at the vessel wall, ensure unimpeded natural convection circulation of the process fluid around the coil perimeter, and provide adequate access for internal inspection and cleaning operations.

Accounting for the 100mm clearance on both sides and the pipe's own outer diameter, the maximum allowable coil centreline diameter is:

$$d_{coil} = d_{tank} - 0.200 - d_{pipe}$$

Where:

- d_{tank} is the internal vessel diameter (m), from Step 3
- 0.200 m accounts for the 100mm radial clearance on each side (2×0.100 m)

- d_{pipe} is the external pipe diameter of the selected NPS (m)

All dimensions must be expressed in consistent units of metres before applying this formula.

4.3.2 Maximum Theoretical Coil Length

The maximum theoretical coil length L_{max} is the total pipe length achievable if the helical coil is wound as tightly as physically possible within the cylindrical shell height h_1 . The minimum structural pitch under tight winding conditions is constrained to twice the external pipe diameter, ensuring that adjacent coil turns do not mechanically clash:

$$p_{min} = 2 \times d_{pipe}$$

The maximum number of turns N_{max} achievable within the cylindrical shell height is:

$$N_{max} = \frac{h_1}{p_{min}}$$

The arc length of a single helical turn is derived from the standard helix geometry, accounting for both the circumferential path length and the vertical rise per turn:

$$L_{turn} = \sqrt{(\pi d_{coil})^2 + (p_{min})^2}$$

The maximum theoretical total coil length is therefore:

$$L_{max} = N_{max} \times L_{turn}$$

$$L_{max} = \frac{h_1}{p_{min}} \times \sqrt{(\pi d_{coil})^2 + (p_{min})^2}$$

The full geometric derivation of the helical arc length formula is provided in Appendix A.

4.3.3 Maximum Theoretical Heat Exchange Surface Area

The external surface area available for heat exchange at maximum theoretical coil length is:

$$A_{max\ theoretical} = \pi \times d_{pip} \times L_{max}$$

This represents the absolute geometric upper limit of the vessel's heat exchange capacity before any real-world manufacturing or operational corrections are applied.

4.4 Packing Efficiency Correction

The maximum theoretical coil length assumes a mathematically perfect helical geometry that is not physically achievable in practice. Manufacturing tolerances, structural support requirements, weld access constraints, thermal expansion allowances, and fluid circulation requirements each consume a portion of the available spatial envelope. The practical coil length $L_{\text{practical}}$ is obtained by applying a compound packing efficiency factor η_{pack} that systematically de-rates the theoretical maximum to a realistic, fabricable boundary.

4.4.1 Packing Efficiency Sub-Factors

The compound packing efficiency factor is the product of five independent sub-factors, each addressing a distinct physical constraint. The designer evaluates each sub-factor against the specific fabrication and operational conditions of the installation and selects the appropriate value from the ranges tabulated below.

Fabrication Quality Factor (η_f)

Accounts for geometric imperfections introduced during pipe bending and helical rolling — including material springback, wall thinning on the outer bend radius, and cross-sectional ovality. These deviations prevent perfect mathematical coil concentricity and require dimensional safety margins to prevent mechanical clashing between adjacent turns during assembly.

Fabrication Quality	η_f
Precision CNC fabrication	0.97–0.99
Industrial standard	0.92–0.96
Low precision/manual	0.85–0.92

Support Structure Factor (η_s)

Accounts for the physical volume consumed by the structural support network required to suspend the coil assembly within the vessel. Under operating conditions, the coil is subjected to its own fluid-filled dead weight, buoyant forces from the surrounding process fluid, and hydrodynamic drag from convective currents. The support skeleton required to resist these loads occupies space and shields portions of the outer pipe perimeter from convective contact with the process fluid.

Support Complexity	η_s
Minimal supports	0.97
Moderate supports	0.93
Heavy support network	0.88

Weld and Cleaning Clearance Factor (η_w)

Accounts for two distinct lifecycle requirements. During fabrication, manual or automated welding torches require a minimum spatial clearance envelope around each joint for torch access, joint execution, and post-weld non-destructive inspection. During operation, the inter-coil spacing must remain sufficient to allow penetration of cleaning solutions — whether manual or automated Clean-In-Place spray systems — to prevent fouling accumulation in confined inter-turn spaces.

Fabrication Access	η_w
Automated TIG	0.98
Standard industrial welding	0.94
Difficult field fabrication	0.88

Thermal Expansion Factor (η_t)

Accounts for the cyclic dimensional growth of the pipe assembly during thermal operation. As the system transitions from ambient standby to operating temperature, the metal undergoes linear and volumetric thermal expansion. Without adequate spatial buffers built into the coil geometry, this expansion generates compressive stress vectors along the pipe axis, causing warping, coil misalignment, and stress corrosion cracking at weld junctions over repeated thermal cycles.

Temperature Severity	η_t
Mild (<50°C)	0.98
Moderate (50–120°C)	0.94
High thermal cycling	0.88

Dead-Zone Prevention Factor (η_d)

Accounts for the fluid dynamics of the process vessel. A coil wound too densely ceases to function as a permeable heat exchange lattice and instead behaves as a solid barrier, choking the natural convective circulation currents that distribute thermal energy through the bulk process fluid. In high-viscosity fluids, this stagnation causes severe localised overheating immediately within the coil interior and leaves the outer bulk fluid underheated, producing dangerous thermal stratification. The dead-zone prevention factor enforces a minimum open flow path through the coil assembly.

Circulation Quality	η_d
Aggressive forced mixing	0.97
Natural convection assisted	0.9
High viscosity stagnant system	0.80–0.88

4.4.2 Compound Packing Efficiency

The five sub-factors are multiplied to yield the compound packing efficiency:

$$\eta_{pack} = \eta_f \times \eta_s \times \eta_w \times \eta_t \times \eta_d$$

The designer should note that the compound factor will typically fall in the range of 0.55 to 0.75 for standard industrial configurations. Values below 0.55 indicate an exceptionally constrained fabrication environment and should prompt a review of the individual sub-factor selections. Values above 0.80 are achievable only under precision CNC fabrication with automated welding and forced fluid mixing, and should not be assumed for general industrial applications.

4.4.3 Maximum Practical Coil Length and Surface Area

The maximum practical coil length is:

$$L_{practical} = L_{max} \times \eta_{pack}$$

The corresponding maximum practical heat exchange surface area is:

$$A_{practical} = \pi \times d_{pipe} \times L_{practical}$$

These values define the upper physical boundary of the coil within the real fabrication and operational constraints of the installation. All thermal performance evaluations from this point forward are bounded by $A_{practical} \cdot L_{max}$ and $A_{max\ theoretical}$ are retained for reference but play no further role in the design calculations.

4.5 Iteration Gate — Geometric Feasibility Check

Before proceeding to pump selection, the designer must confirm that the maximum practical coil configuration is geometrically capable of supporting Q_{design} . This is a preliminary check only — the full thermal performance evaluation is conducted in Steps 6 and 7. The purpose here is to catch a fundamental geometric insufficiency early, before pump selection effort is invested in a configuration that cannot work.

A coil configuration is considered geometrically feasible if the maximum practical surface area $A_{practical}$ is sufficient to deliver Q_{design} under a conservative estimate of the overall heat transfer coefficient U . If:

$$Q_{estimated} = U_{conservative} \times A_{practical} \times \Delta T_{drive}$$

is substantially less than Q_{design} , the current pipe diameter is insufficient. The designer must return to Section 4.2, increment the NPS upward by one standard dimension, and repeat all calculations from that point. The larger pipe

diameter increases d_{coil} , which increases L_{max} and consequently $L_{practical}$, expanding the available heat exchange surface area.

Where the geometric feasibility check passes, the designer proceeds to Step 5 with confidence that the coil envelope is capable of supporting the thermal target, subject to confirmation through the full Effectiveness-NTU analysis in Step 6.

4.6 Step 4 Output Summary

Upon completion of Step 4, the designer has established the following:

- The confirmed pipe material and its engineering justification
- The assigned velocity ceiling v_{max} for the confirmed material-fluid pairing
- The selected Nominal Pipe Size with external diameter d_{pipe} , internal diameter d_{in} , wall thickness t_w , and internal flow area $A_{selected}$
- The actual fluid velocity v_{actual} at the target flow rate, confirmed to be at or below v_{max}
- The maximum coil centreline diameter d_{coil}
- The minimum structural pitch p_{min}
- The maximum theoretical coil length L_{max} and surface area $A_{max\ theoretical}$
- The compound packing efficiency η_{pack} with all five sub-factor values documented
- The maximum practical coil length $L_{practical}$ and surface area $A_{practical}$
- Confirmation that the geometric feasibility check has been passed

These outputs define the physical coil envelope and the hydraulic pipe geometry that govern all subsequent pump selection, thermal performance evaluation, and optimization steps.

Step 5: Hydraulic Resistance Characterisation and Pump Selection

Overview

Step 5 characterises the hydraulic behaviour of the circulation loop and selects a pump capable of driving the thermofluid through the coil assembly at a flow rate that satisfies the thermal design target without breaching the velocity ceiling established in Step 4. This step is fundamentally an interaction problem — the pump and the piping system are not independent components. Their combined behaviour is governed by the intersection of two curves: the system resistance curve, which describes how much pressure the piping network demands at any given flow rate, and the pump performance curve, which describes how much pressure the pump can deliver at any given flow rate. The steady-state operating condition of the circulation loop is defined entirely by this intersection.

The objective of Step 5 is to identify a pump whose natural operating intersection with the system curve produces a flow rate that delivers adequate thermal throughput while maintaining fluid velocity at or below v_{max} under all steady-state conditions. Where the natural intersection violates the velocity ceiling, Variable Frequency Drive scaling is applied to shift the operating point to a compliant condition.

5.1 System Hydraulic Resistance Characterisation

The system resistance curve describes the total pressure loss the circulation pump must overcome as a function of volumetric flow rate. It is composed of two distinct contributions: major losses arising from fluid friction along the pipe length, and minor losses arising from localised flow disturbances at fittings, bends, valves, and manifold connections. Both contributions must be evaluated and summed to construct an accurate system curve.

The system curve is evaluated across a flow rate range spanning from zero to at least 150 percent of the target volumetric flow rate Q_{flow} established in Step 4. Evaluating the curve across this extended range ensures that the pump-system intersection is fully characterised and that the VFD clamping range can be properly defined. A minimum of ten discrete flow rate points distributed across this range is recommended to produce a smooth, well-defined curve.

5.1.1 Major Losses — Helical Coil Friction

Major losses represent the pressure energy dissipated by fluid friction along the internal pipe walls over the full practical coil length $L_{practical}$. In a straight pipe, this is governed directly by the Darcy-Weisbach equation. In a helical coil, the centrifugal force field induced by the pipe curvature generates secondary counter-rotating flow structures — Dean vortices — that increase the effective friction factor relative to a straight pipe of identical length and diameter. This curvature effect must be accounted for explicitly.

Step 5.1.1a — Straight Pipe Friction Factor

The baseline Darcy friction factor for straight pipe flow $f_{straight}$ is determined from the Colebrook-White equation, which accounts for both pipe wall roughness and Reynolds number effects under turbulent flow conditions. The Colebrook-White equation is implicit in $f_{straight}$ and is solved numerically using the Newton-Raphson iterative method seeded with the Haaland approximation. The full numerical solution procedure is provided in Appendix A.

The Reynolds number required for this calculation is:

$$Re = \frac{\rho \times v \times d_{in}}{\mu}$$

Where:

- ρ is the thermofluid density at bulk operating temperature (kg/m^3)
- v is the fluid velocity at the evaluation flow rate (m/s)
- d_{in} is the internal pipe diameter of the selected NPS (m)
- μ is the dynamic viscosity at bulk operating temperature (Pa·s)

The pipe absolute roughness ϵ for the selected pipe material is obtained from Appendix C.

Step 5.1.1b — Dean Number and Flow Regime Classification

The intensity of the secondary flow field induced by helical curvature is quantified by the dimensionless Dean number:

$$De = Re \times \sqrt{\frac{d_{in}}{d_{coil}}}$$

Where d_{coil} is the mean centreline diameter of the helical coil (m), equal to d_{coil} as established in Step 4.

The Dean number governs which of Ito's empirical helical friction factor correlations applies. The flow regime is classified as follows:

- $De < 13.5$ — Curvature effects are negligible. The straight pipe friction factor applies directly: $f_{helical} = f_{straight}$
- $13.5 \leq De \leq 2000$ — Laminar or transitional regime with significant Dean vortex activity. The laminar helical correction applies.
- $De > 2000$ — Turbulent regime. The turbulent helical correction applies.

Step 5.1.1c — Helical Friction Factor Corrections (Ito's Correlations)

For the laminar and transitional regime ($13.5 \leq De \leq 2000$):

$$f_{helical} = f_{straight} \times \frac{21.5 \times De}{(1.56 + \log_{10}(De))^{5.73}}$$

For the turbulent regime ($De \leq 2000$), expressed using the curvature ratio:

$$f_{helical} = f_{straight} \times \left[Re \times \left(\frac{d_{in}}{d_{coil}} \right)^2 \right]^{\frac{1}{20}}$$

The full derivation of Ito's correlations and their applicable Reynolds number boundaries are provided in Appendix A.

Step 5.1.1d — Major Pressure Loss

The major pressure loss across the full practical coil length is:

$$\Delta P_{major} = f_{helical} \times \left(\frac{L_{practical}}{d_{in}} \right) \times \left(\frac{\rho \times v^2}{2} \right)$$

Where all variables are evaluated at the flow rate under consideration. This calculation is repeated at each discrete flow rate point across the evaluation range to construct the major loss component of the system curve.

5.1.2 Minor Losses — Fittings, Valves, and Manifold Connections

Minor losses represent the localised pressure drops induced by flow disturbances at discrete points in the piping network. In extended helical coil systems with multiple inline components, minor losses can contribute between 15 and 25 percent of total system resistance. Omitting them produces a system curve that is systematically optimistic, causing the actual duty point to occur at a higher flow rate — and therefore higher fluid velocity — than the design predicts.

Minor losses are quantified using the K-factor method, where each fitting or flow disturbance is assigned a dimensionless resistance coefficient K representing its pressure loss as a multiple of the dynamic pressure:

$$\Delta P_{minor} = k \times \left(\frac{\rho \times v^2}{2} \right)$$

The total minor loss is the sum of all individual K contributions across the circulation loop:

$$\Delta P_{minor\ total} = \sum k \times \left(\frac{\rho \times v^2}{2} \right)$$

Reference K-factor values for common hydronic loop components are tabulated below. Where the exact fitting geometry is not yet determined at this stage of design, the conservative upper bound value should be applied.

Component	K-Factor Range	Recommended Conservative Value
Gate valve — fully open	0.1 – 0.2	0.2
Globe valve — fully open	6.0 – 10.0	10
Ball valve — fully open	0.05 – 0.15	0.15
90° standard elbow	0.6 – 0.9	0.9
90° long-radius elbow	0.2 – 0.4	0.4
Tee — flow through run	0.3 – 0.6	0.6
Tee — flow through branch	0.9 – 1.8	1.8
Coil inlet / outlet manifold connection	0.5 – 1.5	1.5
Inline strainer — clean condition	1.5 – 3.0	3
Check valve — swing type	1.5 – 2.5	2.5

The designer should compile an itemised schedule of all fittings and inline components in the circulation loop and sum their K contributions. Where the loop configuration is not fully detailed at this stage, a blanket minor loss allowance of 20 percent applied to the major loss value is an acceptable preliminary estimate, to be replaced by an itemised calculation as the detailed design progresses.

$$\Delta P_{minor\ allowance} = 0.20 \times P_{major}$$

5.1.3 Total System Resistance

The total system pressure loss at any given flow rate is the sum of major and minor contributions:

$$\Delta P_{system} = \Delta P_{major} + \Delta P_{minor\ total}$$

For pump selection and system curve construction purposes, pressure loss is commonly expressed as equivalent fluid head H_{system} in metres of fluid column:

$$H_{system} = \frac{\Delta P_{system}}{\rho \times g}$$

Where g is gravitational acceleration (9.81 m/s^2).

This calculation is performed at each discrete flow rate point across the evaluation range. The resulting dataset of (

Q, H_{system}) pairs define the system resistance curve. As fluid velocity — and therefore flow rate — increases, both major and minor losses increase approximately with the square of velocity, producing the characteristic upward-curving parabolic system resistance profile.

5.2 Pump Selection

With the system resistance curve fully characterised, a candidate circulation pump is selected. The pump must be capable of delivering sufficient head to overcome the system resistance at the target flow rate while driving the fluid at a velocity that remains at or below v_{max} .

5.2.1 Pump Selection Criteria

The following criteria govern candidate pump evaluation and must each be assessed before a pump is confirmed:

Hydraulic Coverage

The pump's performance curve — describing the relationship between developed head H_{pump} and volumetric flow rate Q at nominal operating frequency — must intersect the system resistance curve within the designer's target operating range. The intersection point, the duty point, must occur at a flow rate at or above Q_{flow} established in Step 4, and the corresponding fluid velocity must remain at or below v_{max} .

Efficiency at Duty Point

The pump's hydraulic efficiency at the natural duty point should be evaluated against its Best Efficiency Point (BEP). Operating significantly away from the BEP — typically more than 20 percent below on either side — introduces hydraulic instability, elevated vibration, accelerated bearing wear, and increased energy consumption. The selected pump's BEP should align as closely as practicable with the intended operating range.

Net Positive Suction Head Margin

The pump's required Net Positive Suction Head ($NPSH_r$) at the duty point flow rate must be comfortably below the available Net Positive Suction Head ($NPSH_a$) of the installation. A minimum margin of 0.5m above $NPSH_r$ is recommended for standard applications. Insufficient NPSH margin causes cavitation, which damages the pump impeller, degrades hydraulic performance, and introduces severe flow instability into the automation control loop. The thermofluid's vapour pressure at its maximum operating temperature — noted as a downstream constraint in Step 2 — is a direct input to the $NPSH_a$ calculation.

Seal and Materials Compatibility

The pump mechanical seal, casing material, and impeller alloy must be chemically compatible with the selected thermofluid and must be rated for the system's operating temperature and pressure. For thermal oil applications,

high-temperature mechanical seal specifications are mandatory. For food-grade thermofluid applications, wetted components must comply with applicable hygienic material standards.

Motor Sizing and VFD Compatibility

The pump motor must be sized to deliver the required shaft power at the duty point with an appropriate service factor. Where Variable Frequency Drive control is required — as is typically the case in this framework — the motor must be confirmed as inverter-duty rated, with insulated bearings specified where the motor frame size warrants it to prevent bearing current damage under VFD operation.

5.2.2 Manufacturer Performance Curve Evaluation

The manufacturer's pump performance curve provides the head-flow relationship at nominal electrical supply frequency — typically 50Hz or 60Hz depending on the installation region. The designer should obtain and review the full published performance data for the candidate pump, including:

- Head versus flow rate curve at nominal frequency
- Efficiency curve overlaid on the head-flow curve
- $NPSH_r$ curve across the operating flow range
- Power consumption curve across the operating flow range
- Permissible operating range boundaries — minimum and maximum flow rates for continuous stable operation

The system resistance curve constructed in Section 5.1 is overlaid onto the pump performance curve to identify the natural duty point.

5.3 Duty Point Determination

The duty point is the unique steady-state equilibrium condition at which the pump-generated head exactly equals the total system resistance. It is determined by the intersection of the system resistance curve and the pump performance curve.

At flow rates below the duty point, the pump generates more head than the system requires — the excess head accelerates the fluid and increases the flow rate toward the intersection. At flow rates above the duty point, the system demands more head than the pump can provide — the flow decelerates toward the intersection. The duty point is therefore a stable self-correcting equilibrium under constant-speed pump operation.

The intersection coordinates (Q_{duty}, H_{duty}) are identified graphically from the overlaid curves or solved numerically by finding the flow rate at which:

$$H_{pump}(Q) = H_{system}(Q)$$

The fluid velocity corresponding to Q_{duty} within the selected pipe internal diameter is:

$$v_{duty} = \frac{Q_{duty}}{A_{selected}}$$

This velocity is the critical output of the duty point determination and is evaluated against v_{max} in the following section.

5.4 Velocity Constraint Verification and VFD Clamping

5.4.1 Velocity Compliance Check

The duty point velocity v_{duty} is compared against the velocity ceiling v_{max} assigned in Step 4:

Condition A — Velocity Compliant: If $v_{duty} \leq v_{max}$, the natural duty point satisfies the velocity constraint. The system may operate at the natural duty point without intervention. Proceed to Section 5.5.

Condition B — Velocity Non-Compliant: If $v_{duty} > v_{max}$, the natural duty point violates the velocity ceiling. Variable Frequency Drive intervention is mandatory before the system can be commissioned.

5.4.2 VFD Frequency Scaling

When the natural duty point velocity exceeds v_{max} , the pump operating speed is reduced by decreasing the VFD output frequency below its nominal value. Reducing the pump speed shifts the entire pump performance curve downward and to the left in accordance with the pump affinity laws, which govern the relationship between pump speed, flow rate, head, and power:

$$Q \propto N, H \propto N^2, P \propto N^3$$

Where N is the pump rotational speed, directly proportional to the VFD output frequency.

The target VFD frequency f_{VFD} required to clamp the fluid velocity to exactly v_{max} is determined as follows. The target flow rate at the velocity ceiling is:

$$Q_{max\ velocity} = v_{max} \times A_{selected}$$

The required speed ratio relative to nominal is:

$$N_{ratio} = \frac{Q_{max\ velocity}}{Q_{duty}}$$

The target VFD frequency is:

$$f_{VFD} = f_{nominal} \times N_{ratio}$$

Where $f_{nominal}$ is the nominal electrical supply frequency (50Hz or 60Hz).

At this reduced frequency, the scaled pump performance curve is re-evaluated and overlaid onto the system resistance curve to confirm that the new intersection — the VFD-clamped duty point — occurs at or below $Q_{max\ velocity}$. The full derivation of the pump affinity law scaling relationships is provided in Appendix A.

5.4.3 VFD Minimum Head Verification

Reducing the pump speed reduces the available head proportionally to N^2 . The designer must verify that the head delivered by the pump at the VFD-clamped operating condition remains sufficient to overcome the system resistance at $Q_{max\ velocity}$:

$$H_{pump\ scaled} = H_{pump\ nominal} \times N_{ratio}^2$$

If $H_{pump\ scaled} \geq H_{system}(Q_{max\ velocity})$ The VFD-clamped condition is hydraulically viable. Proceed to Section 5.5.

If $H_{pump\ scaled} < H_{system}(Q_{max\ velocity})$, the speed reduction required to satisfy the velocity ceiling degrades the pump head below the level needed to sustain circulation. This condition cannot be resolved by VFD adjustment alone. The iteration gate in Section 5.6 is triggered.

5.5 Confirmed Operating Flow Rate

Upon passing the velocity compliance check — either at the natural duty point or at the VFD-clamped condition — the confirmed steady-state operating flow rate $Q_{operating}$ is recorded. This is the flow rate the circulation loop will sustain under normal operating conditions and is the primary hydraulic input to the thermal performance evaluation in Step 6.

$$Q_{operating} \leq Q_{max\ velocity} \leq v_{max} \times A_{selected}$$

The corresponding confirmed fluid velocity $v_{operating}$ is:

$$v_{operating} = \frac{Q_{operating}}{A_{selected}}$$

Both values are carried forward to Step 6.

5.6 Iteration Gate — Hydraulic Constraint Violations

Two conditions trigger a mandatory return to earlier design steps. The designer must not proceed past this gate until both conditions are satisfied.

Trigger 1 — Velocity Ceiling Unresolvable by VFD

If the VFD minimum head verification in Section 5.4.3 fails — meaning the pump cannot sustain circulation at the velocity-compliant flow rate — the pipe diameter is the bottleneck. The designer returns to Step 4, increments the NPS upward by one standard dimension, and repeats all calculations from that point. The larger internal diameter simultaneously reduces v_{actual} at any given flow rate and reduces the hydraulic resistance of the system, both of which relieve the constraint.

Trigger 2 — System Pressure Drop Exceeds Pump Seal Limits

If the total system pressure loss ΔP_{system} at the operating flow rate exceeds the rated operating pressure of the selected pump's mechanical seal assembly, the hydraulic configuration is unsafe for continuous operation. The designer must modify the piping layout — for example by splitting the single helical coil run into two parallel coil circuits served by a common header — to reduce the total resistance presented to each pump, and re-evaluate the system curve. In both cases, the revised configuration is carried back through Steps 4 and 5 in full before proceeding.

5.7 Step 5 Output Summary

Upon completion of Step 5, the designer has established the following:

- The system resistance curve across the full evaluation flow rate range, comprising major and minor loss contributions at each discrete operating point
- The selected pump with its confirmed performance curve, efficiency profile, NPSH_r data, and materials specification
- The natural duty point coordinates (Q_{duty}, H_{duty})
- The duty point fluid velocity v_{duty} and its compliance status against v_{max}
- Where VFD intervention is required: the target VFD frequency f_{VFD} , the scaled duty point coordinates, and confirmation that adequate pump head is maintained at the clamped condition
- The confirmed steady-state operating flow rate $Q_{operating}$ and fluid velocity $v_{operating}$
- Confirmation that both iteration gate conditions in Section 5.6 are satisfied

These outputs define the confirmed hydraulic operating condition of the circulation loop and serve as the primary inputs to the thermal performance evaluation in Step 6.

Step 6: Thermal Performance Evaluation

Overview

Step 6 evaluates the thermal performance of the coil assembly at the confirmed operating flow rate $Q_{operating}$ established in Step 5. The objective is to determine the maximum thermal power the physical hardware can transmit to the process fluid under both steady-state operating conditions and worst-case cold-start conditions. This maximum transmissible power Q_{max} is then compared against the fixed design thermal target Q_{design} carried forward from Step 1.

This step is where the hydraulic and thermal tracks of the framework converge for the first time. The flow rate is now fixed by hydraulic constraints. The coil geometry is fixed by the vessel envelope and packing efficiency. The only remaining question is whether the combined hardware — at the confirmed flow rate, with the confirmed surface area, under realistic heat transfer conditions — can deliver what the process demands.

Two thermal evaluations are performed in sequence. The first uses the overall heat transfer coefficient evaluated at steady-state bulk operating conditions and establishes the nominal thermal performance of the system. The second uses the worst-case overall heat transfer coefficient evaluated at cold-start viscosity conditions and establishes the thermal performance floor — the minimum transmissible power the system can guarantee at any point during its operating cycle. The coil geometry must satisfy Q_{design} under both conditions.

6.1 Overall Heat Transfer Coefficient

The overall heat transfer coefficient U is the single most influential variable in the thermal performance calculation. It quantifies the total resistance to heat transfer between the thermofluid flowing inside the coil and the process fluid surrounding it outside. A conservative, physically grounded estimate of U is essential — overprediction of U produces an undersized coil that fails to meet the thermal target in service.

U is computed from a series thermal resistance network spanning the internal fluid boundary layer, the pipe wall, the external fluid boundary layer, and the fouling layers that accumulate on both pipe surfaces over time:

$$\frac{1}{U} = \frac{1}{h_{in}} + R_{f\ in} + \frac{\left(d_{out} \times \ln\left(\frac{d_{out}}{d_{in}}\right)\right)}{2 \times k_{pipe}} + R_{f\ out} + \frac{1}{h_{out}}$$

Where:

- h_{in} is the internal convective heat transfer coefficient ($\text{W}/\text{m}^2 \cdot \text{K}$)
- $R_{f\ in}$ is the internal fouling resistance ($\text{m}^2 \cdot \text{K}/\text{W}$)
- $\frac{\left(d_{out} \times \ln\left(\frac{d_{out}}{d_{in}}\right)\right)}{2 \times k_{pipe}}$ is the cylindrical pipe wall conductive resistance per unit area ($\text{m}^2 \cdot \text{K}/\text{W}$)

- $R_{f\ out}$ is the external fouling resistance ($\text{m}^2\cdot\text{K}/\text{W}$)
- h_{out} is the external convective heat transfer coefficient ($\text{W}/\text{m}^2\cdot\text{K}$)

Each component is evaluated in the sections that follow.

6.1.1 Internal Convective Heat Transfer Coefficient

The internal convective heat transfer coefficient h_{in} characterises the rate at which thermal energy is transferred from the bulk thermofluid to the inner pipe wall surface. It is determined from the Nusselt number Nu_{in} , which relates the convective heat transfer performance to the fluid's thermal conductivity:

$$h_{in} = Nu_{in} \times \frac{k_{fluid}}{d_{in}}$$

Where k_{fluid} is the thermal conductivity of the thermofluid at bulk operating temperature ($\text{W}/\text{m}\cdot\text{K}$) and d_{in} is the internal pipe diameter (m).

The Nusselt number for internal flow in a helical coil is elevated relative to straight pipe flow due to the secondary Dean vortex mixing that enhances heat transfer at the pipe wall. The appropriate Nusselt number correlation depends on the flow regime, characterised by the Reynolds number Re and Dean number De computed in Step 5.

Turbulent Flow Regime ($Re > 10,000$)

For fully turbulent flow within the helical coil, the Gnielinski correlation modified for helical geometry is applied:

$$Nu_{in} = \left(\frac{f_{helical}}{8} \right) \times (Re - 1000) \times \frac{Pr}{\left(1 + 2.17 \sqrt{\frac{f_{helical}}{8}} \right)} \times \left(Pr^{\frac{2}{3}} - 1 \right) \times \left(1 + \left(\frac{d_{in}}{d_{coil}} \right)^{0.9} \right)$$

Where:

- $f_{helical}$ is the helical friction factor computed in Step 5 (dimensionless)
- Pr is the Prandtl number of the thermofluid: $Pr = \mu \times \frac{C_p}{k_{fluid}}$ (dimensionless)
- The final curvature correction term $\left(1 + \left(\frac{d_{in}}{d_{coil}} \right)^{0.9} \right)$ accounts for the enhancement of heat transfer due to Dean vortex secondary flow

Laminar and Transitional Flow Regime ($Re \leq 10,000$)

For laminar or transitional flow conditions — common in high-viscosity thermofluid applications at startup — the Sieder-Tate correlation for helical coil geometry is applied:

$$Nu_{in} = 1.86 \times \left(Re \times Pr \times \frac{d_{in}}{L_{practical}} \right)^{\frac{1}{3}} \times \left(\frac{\mu_{bulk}}{\mu_{wall}} \right)^{0.14} \times \left(1 + 3.6 \times \left(1 - \frac{d_{in}}{d_{coil}} \right) \times \left(\frac{d_{in}}{d_{coil}} \right)^{0.8} \right)$$

Where:

- μ_{bulk} is the dynamic viscosity at the bulk fluid temperature (Pa·s)
- μ_{wall} is the dynamic viscosity evaluated at the pipe wall temperature (Pa·s)
- The viscosity ratio $\left(\frac{\mu_{bulk}}{\mu_{wall}} \right)^{0.14}$ accounts for the effect of viscosity variation across the thermal boundary layer

The full derivations of both Nusselt number correlations and their applicable range boundaries are provided in Appendix A.

6.1.2 Pipe Wall Conductive Resistance

The conductive resistance of the pipe wall is evaluated using the standard cylindrical wall conduction formula. For thin-walled pipes where the wall thickness is small relative to the pipe diameter, this simplifies to:

$$R_{wall} = d_{out} \times \frac{\ln\left(\frac{d_{out}}{d_{in}}\right)}{2 \times k_{pipe}}$$

Where k_{pipe} is the thermal conductivity of the pipe material (W/m·K), obtained from Appendix C for the confirmed pipe material. For stainless steel TP316L, $k_{pipe} = 15.0$ W/(m·K). For carbon steel A106, $k_{pipe} = 43.0$ W/(m·K). For copper B88, $k_{pipe} = 401.0$ W/(m·K).

The substantially higher thermal conductivity of copper relative to stainless steel is reflected here — copper's pipe wall resistance is approximately 27 times lower than stainless steel at equivalent geometry. This partially compensates for copper's more restrictive velocity ceiling in applications where heat transfer coefficient is the primary sizing constraint.

6.1.3 Fouling Resistance Values

Fouling resistances account for the insulating deposit layers that accumulate on internal and external pipe surfaces over the operational service life of the coil. These deposits — arising from fluid degradation products, dissolved mineral precipitation, biological growth, or particulate settlement — progressively reduce the effective heat transfer performance of the coil over time. Designing to clean-surface heat transfer coefficients produces a coil that meets its thermal target initially but degrades below it within months of commissioning.

The framework specifies fouling resistances to be applied at the design stage to ensure the coil remains thermally adequate throughout its operational service life, not merely at initial commissioning. The following reference values are provided for common thermofluid and process fluid pairings:

Internal Fouling Resistance $R_{f\,in}$ (Thermofluid Side)

Thermofluid Class	$R_{f\,in}$ (m ² ·K/W)
Treated utility water — inhibited closed loop	0.000088
Ethylene glycol-water solution — inhibited	0.000176
Synthetic thermal oil — clean service	0.000352
Synthetic thermal oil — aged or degraded	0.000528
Food-grade thermal oil — clean service	0.000352

External Fouling Resistance $R_{f\,out}$ (Process Fluid Side)

Process Fluid Class	$R_{f\,out}$ (m ² ·K/W)
Light refined vegetable oils	0.000352
Heavy crude vegetable oils — palm, tallow	0.000528
Petroleum distillates — clean service	0.000176
Heavy hydrocarbons — high fouling potential	0.000704
Aqueous process streams — clean	0.000088
Polymer melts and wax products	0.000528

Where the process fluid or thermofluid does not appear in these tables, the designer should consult the fouling factor data compiled in Perry's Chemical Engineers' Handbook and apply the most conservative applicable value. Under no circumstances should fouling resistances be omitted from the U calculation on the basis that the system will be regularly cleaned.

6.1.4 External Convective Heat Transfer Coefficient

The external convective heat transfer coefficient h_{out} characterises the rate at which thermal energy is transferred from the outer pipe wall surface to the surrounding bulk process fluid. For submerged coil configurations in large process vessels without mechanical agitation, heat transfer on the external surface is governed by natural convection driven by buoyancy forces arising from density differences between the heated fluid layer adjacent to the coil surface and the cooler bulk fluid.

The external Nusselt number for natural convection from a horizontal cylinder — a standard approximation for individual coil turns — is given by the Churchill-Chu correlation:

$$Nu_{out} = \left\{ 0.6 + \frac{\left[0.387 \times Ra^{\frac{1}{4}} \right]}{\left[1 + \left(\frac{0.559}{Pr} \right)^{\left(\frac{9}{16} \right)} \right]^{\left(\frac{8}{27} \right)}} \right\}^2$$

Where the Rayleigh number Ra is:

$$Ra = \frac{(g \times \beta \times \Delta T_{drive} \times d_{out}^3)}{(v \times \alpha)}$$

And:

- g is gravitational acceleration (9.81 m/s^2)
- β is the thermal expansion coefficient of the process fluid ($1/\text{K}$)
- ΔT_{drive} is the temperature difference between the outer pipe wall and the bulk process fluid (K)
- d_{out} is the external pipe diameter (m)
- v is the kinematic viscosity of the process fluid (m^2/s)
- α is the thermal diffusivity of the process fluid (m^2/s)

The external heat transfer coefficient is then:

$$h_{out} = Nu_{out} \times \frac{K_{process}}{d_{out}}$$

Where $K_{process}$ is the thermal conductivity of the process fluid ($\text{W/m}\cdot\text{K}$). For installations with mechanical agitation or forced circulation of the process fluid, h_{out} will be substantially higher than the natural convection estimate. The natural convection value is retained as the conservative baseline in all cases where forced mixing cannot be guaranteed throughout the full operating cycle. The full derivation of the Churchill-Chu correlation and the Rayleigh number formulation are provided in Appendix A.

6.2 Steady-State Overall Heat Transfer Coefficient

With all resistance components evaluated, the steady-state overall heat transfer coefficient U_{steady} is assembled from the series resistance network:

$$\frac{1}{U_{steady}} = \frac{1}{h_{in steady}} + R_{f in} + R_{wall} + R_{f out} + \frac{1}{h_{out steady}}$$

Where $h_{in steady}$ and $h_{out steady}$ are evaluated using fluid properties at the bulk operating temperature established in Step 2.

U_{steady} represents the nominal heat transfer performance of the coil under stabilised operating conditions and is used in the steady-state thermal performance evaluation.

6.3 Worst-Case Overall Heat Transfer Coefficient

The worst-case overall heat transfer coefficient U_{cold} accounts for the most thermally demanding condition the system encounters during normal operation — the cold-start phase, when the thermofluid has not yet reached its operating temperature and its viscosity is at its maximum value.

The cold-start viscosity μ_{cold} flagged in Step 2 is substituted into the internal Nusselt number correlation in place of μ_{bulk} . All other fluid properties are evaluated at the cold-start temperature. The resulting internal heat transfer coefficient $h_{in cold}$ will be substantially lower than $h_{in steady}$ for viscous thermofluid classes, reflecting the degraded boundary layer mixing at high viscosity conditions.

$$\frac{1}{U_{cold}} = \frac{1}{h_{in cold}} + R_{f in} + R_{wall} + R_{f out} + \frac{1}{h_{out cold}}$$

U_{cold} is the conservative design value used for coil surface area sizing. Designing to U_{cold} ensures that the coil meets its thermal target even during the most demanding phase of operation, not merely at stabilised steady-state conditions.

6.4 Logarithmic Mean Temperature Difference

The logarithmic mean temperature difference ΔT_{lm} represents the effective thermal driving force between the thermofluid and the process fluid across the full length of the coil. For a submerged coil operating within a large, well-mixed process vessel, the process fluid temperature T_{tank} is approximately uniform throughout the vessel volume. Under this condition, the LMTD simplifies considerably.

The thermofluid enters the coil at temperature $T_{in} = T_{supply}$ and exits at temperature $T_{out} = T_{supply} - \Delta T_{coil}$

The process fluid remains at approximately T_{tank} throughout. The temperature differences at the coil inlet and outlet are therefore:

$$\Delta T_1 = T_{in} - T_{tank} = T_{supply} - T_{tank}$$

$$\Delta T_2 = T_{out} - T_{tank} = (T_{supply} - \Delta T_{coil}) - T_{tank}$$

The logarithmic mean temperature difference is:

$$\Delta T_{lm} = \frac{(\Delta T_1 - \Delta T_2)}{\ln\left(\frac{\Delta T_1}{\Delta T_2}\right)}$$

Where $\ln$ denotes the natural logarithm. If ΔT_1 and ΔT_2 are within 5 percent of each other, the arithmetic mean may be used in place of the logarithmic mean without meaningful loss of accuracy.

The designer should note that ΔT_{lm} is sensitive to the thermofluid supply temperature T_{supply} . Higher supply temperatures increase the driving force and reduce the required coil surface area. However T_{supply} is bounded above by the thermofluid's rated thermal stability limit and below by the requirement that T_{supply} must exceed T_{tank} by a meaningful margin for heat transfer to occur in the correct direction.

6.5 Thermal Performance Evaluation — Effectiveness-NTU Method

The thermal performance of the coil assembly is evaluated using the Effectiveness-NTU method. This approach is selected over the direct LMTD method because it does not require prior knowledge of the thermofluid outlet temperature — a parameter that is determined by the coil's heat exchange performance rather than being independently specified.

6.5.1 Thermal Capacity Rate and Maximum Potential Heat Transfer

The thermal capacity rate of the thermofluid circulating through the coil is:

$$C_{coil} = \dot{m} \times C_p = (Q_{operating} \times \rho) \times C_p$$

Where $\dot{m}$ is the thermofluid mass flow rate (kg/s).

For a submerged coil in a large well-mixed process vessel, the thermal capacity rate of the process fluid C_{tank} approaches infinity relative to C_{coil} as the bulk tank temperature remains essentially constant regardless of the heat input from the coil during any incremental time step. Consequently:

$$C_{min} = C_{coil}$$

$$C_{max} = C_{tank} \rightarrow \infty$$

$$C_r = \frac{C_{min}}{C_{max}} = 0$$

The maximum theoretically achievable heat transfer rate — if the thermofluid were brought entirely to the process fluid temperature over the full coil length — is:

$$Q_{fluid} = C_{min} \times |T_{supply} - T_{tank}| = \dot{m} \times C_p \times (T_{supply} - T_{tank})$$

6.5.2 Number of Transfer Units

The Number of Transfer Units NTU quantifies the thermal sizing of the heat exchanger relative to the fluid's heat capacity rate. For $C_r = 0$, the NTU-effectiveness relationship simplifies to:

$$\varepsilon = 1 - \exp(-NTU)$$

Where heat exchanger effectiveness ε is defined as the ratio of actual heat transfer to the thermodynamic maximum:

$$\varepsilon = \frac{Q_{actual}}{Q_{fluid}}$$

The NTU is defined as:

$$NTU = U \times \frac{A_s}{C_{min}}$$

Where A_s is the coil external heat exchange surface area.

6.5.3 Maximum Transmissible Thermal Power

Combining the NTU formulation with the effectiveness relationship, the maximum thermal power transmissible by the coil assembly at the confirmed operating condition is:

$$Q_{max} = Q_{fluid} \times (1 - \exp(-NTU))$$

$$Q_{max} = \dot{m} \times C_p \times (T_{supply} - T_{tank}) \times \left(1 - \exp\left(\frac{-U \times A_s}{\dot{m} \times C_p}\right)\right)$$

This is evaluated twice — once using U_{steady} and $A_{practical}$ to establish the nominal steady-state maximum, and

once using U_{cold} and $A_{practical}$ to establish the worst-case cold-start maximum.

The full derivation of the Effectiveness-NTU method and the simplification for $C_r = 0$ are provided in Appendix A.

6.6 Thermal Adequacy Verification

The maximum transmissible thermal power under both operating conditions is compared against the fixed design thermal target Q_{design} established in Step 1:

Condition A — Thermally Adequate: If $Q_{max} \geq Q_{design}$ under both steady-state and cold-start conditions, the coil configuration is thermally adequate. The hardware can deliver the required thermal power with the factor of safety already embedded in Q_{design} . Proceed to Step 7.

Condition B — Thermally Inadequate at Cold-Start Only: If Q_{max} under cold-start conditions falls below Q_{design} but Q_{max} under steady-state conditions satisfies it, the cold-start performance is the binding constraint. The designer should consider increasing the thermofluid supply temperature during startup, specifying a thermofluid pre-heating arrangement, or increasing the coil surface area. If no adjustment is practicable, this condition must be formally documented as an operational limitation and the startup procedure must be defined accordingly.

Condition C — Thermally Inadequate Under Both Conditions: If Q_{max} falls below Q_{design} under both evaluation conditions, the coil configuration is fundamentally insufficient. The designer returns to Step 4, increments the NPS upward by one standard dimension, and repeats Steps 4, 5, and 6 in full. The larger pipe diameter increases d_{coil} , expands $L_{practical}$, and increases A_s — all of which improve Q_{max} .

6.7 Step 6 Output Summary

Upon completion of Step 6, the designer has established the following:

- The steady-state overall heat transfer coefficient U_{steady} with all resistance components itemised
- The worst-case cold-start overall heat transfer coefficient U_{cold}
- The logarithmic mean temperature difference ΔT_{lm} at the confirmed operating condition
- The thermal capacity rate C_{coil} and the maximum potential heat transfer Q_{fluid}
- The Number of Transfer Units NTU at maximum practical coil surface area
- The maximum transmissible thermal power Q_{max} under steady-state and cold-start conditions
- Confirmation that $Q_{max} \geq Q_{design}$ under both evaluation conditions, or a documented resolution of any deficiency identified

These outputs confirm the thermal viability of the coil configuration and serve as the primary inputs to the geometric optimization in Step 7.

Step 7: Coil Geometry Optimisation

Overview

Step 7 scales the coil configuration down from its maximum practical geometry to the minimum physically realisable configuration that satisfies the fixed design thermal target Q_{design} . The maximum practical coil established in Step 4 represents the upper physical boundary of the vessel's heat exchange capacity. Operating a system at its absolute geometric ceiling is rarely economically justified — it produces unnecessary material cost, elevated hydraulic resistance, extended fabrication lead times, and a system that is significantly more powerful than the process requires.

The optimisation procedure begins at $L_{practical}$ and systematically reduces the coil length until the transmissible thermal power Q_{max} converges to Q_{design} . The factor of safety embedded in Q_{design} from Step 1 means the optimised coil is not the thermodynamic minimum — it is the safety-margined minimum. The coil is sized to reliably deliver the required process heat load under fouled, cold-start, and degraded operating conditions, not merely under ideal clean-surface steady-state conditions.

Following coil length optimization, the hydraulic implications of the shortened coil are evaluated and the pump selection from Step 5 is reviewed to determine whether a reduced pump specification is warranted.

7.1 Optimisation Objective and Stopping Criterion

The optimisation objective is to find the minimum coil length L_{coil} such that the maximum transmissible thermal power Q_{max} at that length equals the design thermal target:

$$Q_{max}(L_{coil}) = Q_{design}$$

The stopping criterion is:

$$\frac{Q_{max}}{Q_{design}} = 1.0 \pm tolerance$$

A convergence tolerance of 2 percent is recommended for preliminary design — that is, the optimisation is considered converged when:

$$0.98 \leq \frac{Q_{max}}{Q_{design}} \leq 1.02$$

Values below the lower bound indicate an undersized coil that will not reliably meet the thermal target. Values above the upper bound indicate residual oversizing that can be further reduced. The designer should always err toward the upper bound of the tolerance band rather than the lower, accepting marginal oversizing in preference to marginal undersizing.

Q_{design} is carried forward unchanged from Step 1. It is not re-derived, adjusted, or replaced at this stage.

7.2 Required Heat Exchange Surface Area

The minimum heat exchange surface area required to deliver Q_{design} is back-calculated from the Effectiveness-NTU framework established in Step 6. Starting from the target effectiveness:

$$\varepsilon_{target} = \frac{Q_{design}}{Q_{fluid}}$$

Where $Q_{fluid} = \dot{m} \times C_p \times (T_{supply} - T_{tank})$ is the maximum potential heat transfer established in Step 6.

The designer should verify that ε_{target} is strictly bounded between 0 and 1. An effectiveness value at or above 1.0 is physically impossible and indicates that the thermofluid cannot carry sufficient thermal energy to meet Q_{design} at the confirmed flow rate — the flow rate or supply temperature must be increased before proceeding. An effectiveness value above 0.9 is theoretically valid but requires exponentially increasing surface area for marginal gains and should prompt the designer to review whether the thermofluid supply temperature can be increased to reduce the required effectiveness.

From the inverted NTU-effectiveness relationship for $C_r = 0$:

$$NTU_{required} = -\ln(1 - \varepsilon_{target})$$

The required coil external surface area is then:

$$A_{required} = NTU_{required} \times \frac{C_{min}}{U_{cold}}$$

Where U_{cold} is the worst-case cold-start overall heat transfer coefficient from Step 6. Using U_{cold} rather than U_{steady} for surface area sizing ensures the coil meets Q_{design} under the most thermally demanding operating condition, not merely at stabilised steady-state conditions.

7.3 Optimised Coil Length

The optimised coil length L_{coil} is derived directly from the required surface area and the external pipe diameter:

$$L_{coil} = \frac{A_{required}}{\pi \times d_{pipe}}$$

This is the minimum pipe length that provides sufficient external heat exchange surface area to satisfy Q_{design} under worst-case thermal conditions.

L_{coil} must satisfy the following boundary condition before the geometry is accepted:

$$L_{coil} \leq L_{practical}$$

If L_{coil} exceeds $L_{practical}$, the maximum practical coil configuration cannot deliver Q_{design} . This is the thermal deficiency condition. The designer must return to Step 4, increment the NPS upward by one standard dimension, and repeat Steps 4 through 7. The larger pipe diameter expands the coil envelope, increases $L_{practical}$, and provides the additional surface area required.

If L_{coil} is less than or equal to $L_{practical}$, the optimised length is geometrically realisable within the vessel envelope. Proceed to Section 7.4.

7.4 Optimised Helical Pitch

Reducing the coil length from $L_{practical}$ to L_{coil} while keeping the coil centreline diameter d_{coil} and the cylindrical shell height h_1 fixed requires the helical pitch to be expanded from its minimum value p_{min} . The number of turns in the optimised coil is reduced, and the vertical spacing between adjacent turns increases accordingly.

The optimised pitch p_{coil} is back-calculated by inverting the helix arc length formula. Given that the total coil length and the coil centreline diameter are known, the pitch is:

$$p_{coil} = \frac{\pi d_{coil}}{\sqrt{\left(\frac{L_{coil}}{N_{coil}}\right)^2 - \left(\frac{\pi d_{coil}}{p_{coil\ approx}}\right)^2}}$$

This expression is implicit and is most practically resolved using the following direct formulation derived from the helical geometry. Given the total coil length L_{coil} wound over a cylindrical height h_1 with coil diameter d_{coil} , the pitch satisfies:

$$L_{coil} = \left(\frac{h_1}{p_{coil}} \right) \times \sqrt{(\pi d_{coil})^2 + (p_{coil})^2}$$

This transcendental equation is solved iteratively for p_{coil} . A Newton-Raphson procedure or simple bisection between p_{min} and h_1 will converge rapidly. The full numerical solution procedure is provided in Appendix A.

7.4.1 Pitch Sanity Check — Minimum Pitch Boundary

The back-calculated pitch p_{coil} must satisfy:

$$p_{coil} \geq p_{min} = 2 \times d_{pipe}$$

If the back-calculated pitch falls below p_{min} , the optimised coil length requires more turns than can be physically accommodated within h_1 at the minimum allowable pitch spacing. This condition means the coil is already at its densest achievable packing and cannot be further reduced without mechanical clashing between adjacent turns.

In this case two options are available. The designer may accept the maximum practical coil configuration $L_{practical}$ as the final coil length, acknowledging that the system will operate with thermal capacity modestly above Q_{design} — a condition that is thermally conservative and operationally acceptable. Alternatively, if the excess capacity is substantial, the designer may return to Step 3 and reduce the vessel height h_1 , which reduces $L_{practical}$ and forces the pitch constraint to be satisfied at a shorter coil length. The implications of reducing h_1 on vessel volume must be assessed before this option is pursued.

7.4.2 Pitch Sanity Check — Maximum Pitch Boundary

The back-calculated pitch p_{coil} must also satisfy an upper bound. If the optimised coil requires so few turns that the pitch becomes excessively large relative to the vessel height, the coil occupies only a small fraction of the available cylindrical shell height and the remaining vessel volume above and below the coil is thermally inactive. A practical upper bound of:

$$P_{coil} \leq \frac{h_1}{3}$$

is recommended, ensuring a minimum of three complete helical turns within the cylindrical shell height. Fewer than three turns produces a coil that is geometrically closer to a flat spiral than a helix and behaves differently from the helical flow assumptions underlying the hydraulic and thermal calculations. If the optimised coil requires fewer than three turns, the designer should reduce d_{coil} by selecting a smaller NPS in Step 4, which reduces L_{coil} and increases the number of turns required for the same surface area.

7.5 Final Coil Turn Count and Height Verification

With p_{coil} confirmed within its sanity bounds, the final number of complete helical turns in the optimised coil is:

$$N_{coil} = \frac{h_1}{p_{coil}}$$

This value should be rounded down to the nearest integer to ensure the coil fits within the cylindrical shell height without encroaching on the cap regions. The actual coil winding height consumed by the integer number of turns is:

$$h_{coil} = N_{coil} \times p_{coil}$$

The remaining clearance between the top of the coil and the base of the upper cap, and between the bottom of the coil and the base of the lower cap, is:

$$h_{clearance} = h_1 - h_{coil}$$

A minimum clearance of 50mm at each end is recommended to ensure adequate fluid circulation at the top and bottom of the coil bundle and to provide access for coil support attachment points. If $h_{clearance}$ falls below this threshold, N_{coil} should be reduced by one turn and the resulting L_{coil} and $A_{required}$ rechecked for compliance with Q_{design} .

The finalised coil specification — L_{coil} , d_{coil} , p_{coil} , N_{coil} , h_{coil} — constitutes the buildable coil geometry and is the primary output of Step 7.

7.6 Pump Review Following Coil Optimisation

Reducing the coil length from $L_{Practical}$ to L_{coil} reduces the total pipe length through which hydraulic friction losses accumulate. This shifts the system resistance curve downward — for any given flow rate, the system now demands less head than it did at maximum coil length. The natural duty point migrates to a higher flow rate and lower head on the pump performance curve.

The designer must re-evaluate the system curve at L_{coil} and confirm the following:

7.6.1 Velocity Re-verification

The new duty point flow rate $Q_{duty\,optimised}$ must be re-checked against the velocity ceiling:

$$v_{optimised} = \frac{Q_{duty\,optimised}}{A_{selected}} \leq v_{max}$$

If the shortened coil has shifted the duty point to a flow rate that pushes the velocity above v_{max} , the VFD clamp frequency must be recalculated using the procedure established in Step 5, Section 5.4.2. The confirmed operating flow rate $Q_{operating}$ is updated accordingly.

7.6.2 Thermal Re-verification at Updated Flow Rate

If $Q_{operating}$ has changed following the pump review, the thermal performance evaluation must be re-confirmed. The updated flow rate alters C_{min} , which affects both Q_{fluid} and the NTU calculation. The designer re-evaluates:

$$Q_{max\,updated} = \dot{m}_{updated} \times c_p \times (T_{supply} - T_{tank}) \times \left(1 - \exp\left(\frac{-U_{cold} \times A_{coil}}{\dot{m}_{updated} \times c_p}\right)\right)$$

Where $A_{coil} = \pi d_{pipe} \times L_{coil}$ is the surface area of the optimised coil. If $Q_{max\,updated} \geq Q_{design}$ the thermal target remains satisfied and the design is confirmed. If $Q_{max\,updated}$ falls below Q_{design} due to a reduction in flow rate caused by the system curve shift, L_{coil} must be marginally increased to compensate.

7.6.3 Pump Downsizing Assessment

Where the shortened coil has substantially reduced the system resistance, the original pump selection from Step 5 may represent significant oversizing in terms of head capacity. The designer should assess whether a smaller pump — with a lower head rating but the same or similar flow range — provides a better fit to the optimised system curve at lower capital cost and improved operating efficiency.

A pump change is warranted when the original pump's duty point on the optimised system curve falls more than 25 percent to the right of its Best Efficiency Point, indicating operation in an inefficient and potentially unstable hydraulic region. Where a smaller pump is selected, its performance curve must be fully re-evaluated against the optimised system curve and all velocity and head checks repeated before the selection is confirmed.

Where the original pump remains appropriate, no further action is required and the pump selection from Step 5 is carried forward unchanged.

7.7 Step 7 Output Summary

Upon completion of Step 7, the designer has established the following:

- The required heat exchange surface area $A_{required}$ derived from Q_{design} and U_{cold}
- The optimised coil length L_{coil} confirmed to satisfy $L_{coil} \leq L_{practical}$
- The back-calculated optimised helical pitch p_{coil} confirmed within minimum and maximum pitch bounds
- The final turn count N_{coil} and winding height h_{coil} confirmed within the cylindrical shell height h_1
- The end clearance $h_{clearance}$ confirmed above the 50mm minimum threshold at each end
- The finalised buildable coil specification: L_{coil} , d_{coil} , p_{coil} , N_{coil} , h_{coil}
- The updated system curve at L_{coil} and confirmation of velocity compliance at the revised duty point
- The thermal re-verification at the updated operating flow rate confirming $Q_{max\ updated} \geq Q_{design}$
- The pump review outcome — either confirmation of the original pump selection or documentation of a revised pump specification

These outputs constitute the complete, optimised, buildable internal coil specification and the confirmed hydraulic operating condition of the final system configuration. They serve as the primary geometric inputs to the volumetric displacement compensation in Step 8.

Step 8: Volumetric Displacement Compensation and Final Vessel Dimensioning

Overview

Step 8 addresses a systematic error that is consistently overlooked in standard hydronic loop design practice. When a helical coil assembly is submerged inside a process vessel, it physically occupies volume that would otherwise be available to the process fluid. A vessel fabricated to its baseline internal dimensions — as established in Step 3 — will therefore deliver less usable process capacity than the client specified once the coil assembly is installed.

For small coil assemblies in large vessels this displacement may be negligible. For extensively wound coils in moderately sized vessels it can represent a meaningful fraction of the total vessel volume — sufficient to cause inventory errors in precision batch processing, automated dosing systems, and blending operations where accurate volume tracking is operationally critical.

The correction applied in this step is conceptually straightforward: the volume displaced by the submerged assembly is quantified, added to the baseline target volume, and the vessel dimensions are adjusted to recover the lost capacity. Critically, the coil geometry established in Step 7 is considered finalised and is not revisited at this stage. The vessel geometry adjusts to accommodate the coil — not the other way around.

8.1 Coil Assembly Displacement Volume

The submerged assembly displaces process fluid equal to the total volume of all solid and fluid-filled material occupying space within the vessel. This includes the coil pipe itself, the fluid contained within the coil, support structures, manifold headers, and ancillary connection components.

8.1.1 Primary Coil Pipe Displacement

The dominant displacement contribution is the coil pipe. For displacement purposes, the entire external envelope of the pipe — including the fluid contained within its bore — is treated as a solid cylinder. This is correct because the fluid inside the coil is thermofluid, not process fluid. It contributes no usable process volume and must be fully accounted for in the displacement calculation.

Using the external pipe diameter d_{pipe} and the optimised coil length L_{coil} from Step 7:

$$V_{coil} = \frac{\pi \times d_{pipe}^2}{4} \times L_{coil}$$

This formulation correctly uses the external pipe diameter. Using the internal diameter would underestimate displacement by excluding the pipe wall volume and is a common error that must be avoided.

8.1.2 Ancillary Assembly Displacement

The support structure network, inlet and outlet manifold headers, connection spools, and structural gussets that form the complete submerged assembly collectively displace additional process fluid volume beyond the coil pipe alone. These components are not individually detailed at this stage of the design process. A conservative volumetric allowance is applied as a percentage of the primary coil displacement:

$$V_{ancillary} = k_{ancillary} \times V_{coil}$$

Where $k_{ancillary}$ is the ancillary displacement factor, selected from the following reference values based on the support complexity tier established in Step 4:

Support Complexity	k_ancillary
Minimal support — lightweight coil, low process fluid density	0.08
Moderate support — standard industrial configuration	0.12
Heavy support network — large coil, high-density process fluid	0.18

The upper bound of each tier is applied in all cases to maintain conservatism. This allowance is replaced by a

precise itemised displacement calculation once the structural support design is finalised.

8.1.3 Total Assembly Displacement Volume

The total volumetric displacement of the complete submerged assembly is:

$$V_{displacement} = V_{coil} + V_{ancillary} = V_{coil} \times (1 + k_{ancillary})$$

8.2 Gross Fabrication Volume Target

To recover the displaced process fluid capacity, the baseline vessel volume V_{target} specified by the client must be augmented by the total assembly displacement:

$$V_{gross} = V_{target} + V_{displacement}$$

V_{gross} is the internal volume the fabricated vessel must provide to ensure that the net usable process fluid capacity — after the coil assembly is installed — precisely meets the client's original specification. The designer must fabricate to V_{gross} , not to V_{target} .

8.3 Dimensional Adjustment Strategy

The designer must satisfy V_{gross} by adjusting one of two primary geometric variables of the vessel established in Step 3. The coil geometry is fixed and is not adjusted. The choice between the two adjustment options is governed by the site constraints and manufacturing considerations of the specific installation.

8.3.1 Option A — Vertical Shell Extension (Preferred Default)

Extending the cylindrical shell height h_1 increases the vessel volume while preserving the internal diameter d_{tank} and all radial clearances unchanged. Because the coil diameter d_{coil} and the radial clearance between the coil and the vessel wall are both functions of d_{tank} — which remains unchanged — the coil assembly fits identically within the extended vessel without any geometric interference.

This option is the preferred default for the following reasons. The radial clearance geometry is preserved exactly. The coil support attachment points on the vessel wall remain at the same radial positions. The vessel footprint on the plant floor does not change. Fabrication of an extended cylindrical shell is simpler and less costly than reforming the vessel diameter. The only constraint that limits this option is available headroom — if the plant ceiling height restricts the total vessel height, Option B must be considered.

The required additional shell height Δh_1 to recover $V_{displacement}$ is derived from the cylindrical volume

relationship:

$$\Delta h_1 = \frac{V_{displacement}}{\left(\frac{\pi d_{tank}^2}{4}\right)}$$

The adjusted total cylindrical shell height is:

$$h_{1\ final} = h_1 + \Delta h_1$$

The total vessel height is updated accordingly:

$$h_{final} = 2 \times h_{1\ final} + h_2$$

Where h_2 is the cap height from Step 3, which remains unchanged.

8.3.2 Option B — Radial Diameter Expansion

Increasing the internal vessel diameter d_{tank} while preserving the cylindrical shell height h_1 also recovers the displaced volume. However this option introduces a geometric consequence that must be acknowledged: increasing d_{tank} widens the radial gap between the outer edge of the coil and the internal vessel wall beyond the 100mm design minimum established in Step 4.

This increased clearance is not harmful — the minimum clearance requirement remains satisfied with margin — but the designer should note that the wider gap may slightly reduce natural convection effectiveness at the vessel wall boundary for high-viscosity process fluids. This effect is minor in most applications and does not warrant a thermal recalculation.

The coil geometry is not adjusted in response to the diameter increase. The coil was finalised in Step 7 and remains unchanged. The vessel simply becomes marginally wider around it.

The adjusted internal diameter $d_{tank\ final}$ is determined by solving the vessel volume formula for the selected cap geometry with V_{gross} substituted for V_{target} . The algebraic solution procedure mirrors that applied in Step 3 and is referenced in Appendix A.

This option is selected only when vertical extension is precluded by headroom constraints or where a wider vessel diameter is operationally advantageous — for example, to improve internal inspection access or accommodate a larger agitator assembly.

8.4 Updated External Surface Area and Q_{loss} Feedback Check

Adjusting the vessel dimensions changes its external surface area relative to the baseline computed in Step 3. The updated external surface area $A_{total\ final}$ must be recomputed using the adjusted dimensions and compared against the baseline value used in the Q_{loss} placeholder estimate in Step 1.

For Option A — vertical shell extension:

$$A_{shell\ final} = \pi \times d_{tank} \times h_{1\ final}$$

$$A_{total\ final} = A_{shell\ final} + A_{caps}$$

Where A_{caps} is unchanged from Step 3.

For Option B — radial diameter expansion:

Both the shell area and the cap areas change with $d_{tank\ final}$ and must be fully recomputed using the surface area expressions established in Step 3.

The updated heat loss placeholder is re-estimated using $A_{total\ final}$ in place of the original A_{total} :

$$Q_{loss\ updated} = k \times m_{product} \times C_{p\ liquid} \times T_{operating} \times \left(\frac{A_{total\ final}}{A_{total}} \right)$$

Where the ratio $\left(\frac{A_{total\ final}}{A_{total}} \right)$ scales the original placeholder proportionally to the change in surface area.

If $Q_{loss\ updated}$ exceeds $Q_{loss\ original}$ by more than 10 percent, the designer must return to Step 1 and recalculate $Q_{balance}$ and Q_{design} with the updated loss estimate before proceeding. A surface area increase of less than 10 percent is considered within the conservatism already embedded in the upper-bound insulation tier values applied in Step 1 and does not require a design loop return.

This check formally closes the Q_{loss} feedback loop introduced in Step 1 and revisited in Step 3. At this point in the design sequence, the vessel geometry is sufficiently well defined that the Q_{loss} estimate is no longer a preliminary placeholder but a defensible engineering approximation, valid until replaced by the rigorous thermal resistance network calculation in the thermal driver sizing step.

8.5 Final Vessel Specification

Upon confirming that the Q_{loss} feedback check is satisfied, the final vessel dimensions are recorded as the buildable fabrication specification. These dimensions supersede the baseline dimensions established in Step 3 in all subsequent steps.

The final vessel specification comprises:

- Internal diameter: d_{tank} (unchanged for Option A) or $d_{tank\ final}$ (Option B)
- Cylindrical shell height: $h_{1\ final}$ (Option A) or h_1 (Option B)
- Cap height: h_2 (unchanged in both options)
- Total vessel height: h_{final}
- Cap profile: as selected in Step 3
- Gross internal volume: V_{gross}
- Net usable process volume: V_{target} (confirmed recoverable after coil assembly installation)
- Updated external surface area: $A_{total\ final}$

8.6 Step 8 Output Summary

Upon completion of Step 8, the designer has established the following:

- The primary coil pipe displacement volume V_{coil} using the external pipe diameter and optimised coil length
- The ancillary assembly displacement allowance $V_{ancillary}$ based on the support complexity tier
- The total assembly displacement volume $V_{displacement}$
- The gross fabrication volume target V_{gross}
- The selected dimensional adjustment option and its engineering justification
- The final vessel dimensions — d_{tank} , $h_{1\ final}$, h_2 , h_{final} — constituting the buildable fabrication specification
- The updated external surface area $A_{total\ final}$
- Confirmation that the Q_{loss} feedback check has been satisfied or documentation of a design loop return to Step 1 where the 10 percent threshold has been exceeded

These outputs define the finalised process vessel geometry. Together with the coil specification from Step 7, they constitute a complete, buildable primary asset specification ready for detailed structural and fabrication engineering.

Step 9: Thermal Driver Sizing

Overview

Step 9 determines the total thermal power capacity required of the primary heating or cooling source — the thermal driver — that supplies conditioned thermofluid to the circulation loop. It also retires the Q_{loss} placeholder estimates carried forward from Steps 1, 3, and 8, replacing them with a rigorous thermal resistance network calculation using the finalised vessel geometry established in Step 8.

Two outputs are produced. The first is the precise steady-state environmental heat loss rate Q_{loss} for each vessel in the installation, computed from first principles using the actual insulation specification and worst-case ambient boundary conditions. The second is the total thermal driver capacity Q_{total} required to simultaneously satisfy the process heat demands and the standing environmental losses of all vessels in the installation under peak loading conditions.

This step also formally resolves the Q_{loss} feedback loop that has been carried open since Step 1. If the rigorously calculated Q_{loss} deviates from the Step 8 updated placeholder by more than the conservatism embedded in the design factor of safety, the framework returns to Step 1 for recalculation. If the deviation is within that margin, the design proceeds without modification.

9.1 Insulation Specification

In Steps 1, 3, and 8 the environmental heat loss was estimated using a placeholder percentage based on an insulation tier selection. Step 9 replaces that placeholder with a real material specification. The designer selects the insulation material, confirms its thermal conductivity, and specifies the insulation layer thickness to be applied to the vessel external surface.

The insulation specification must be consistent with the tier selected in Step 1. A designer who selected the Well Insulated tier in Step 1 must specify an insulation system that genuinely achieves that performance level. Specifying a thinner or lower-conductivity insulation than the tier implies would invalidate the conservatism of the Q_{loss} estimate used throughout the earlier steps.

Common industrial insulation materials and their thermal conductivity ranges at typical operating temperatures are:

Insulation Material	Thermal Conductivity k_{ins} (W/m·K)	Typical Application
Mineral wool — glass fibre	0.033 – 0.045	General industrial, up to 250°C
Mineral wool — rock wool	0.035 – 0.050	High temperature, up to 400°C
Polyurethane foam — rigid	0.022 – 0.028	Ambient to moderate temperature, up to 120°C
Cellular glass	0.038 – 0.055	Cryogenic to high temperature
Calcium silicate	0.055 – 0.075	High temperature industrial, up to 650°C
Aerogel blanket	0.012 – 0.018	High performance, space-constrained installations

The thermal conductivity value used in all subsequent calculations must be the upper bound of the selected material's range at the operating temperature of the installation. This maintains conservatism — a higher thermal conductivity produces a higher Q_{loss} estimate, which results in a more conservatively sized thermal driver.

The insulation layer thickness L_{ins} is specified by the designer based on the installation requirements, available space, and economic considerations. Thicker insulation reduces Q_{loss} but increases material cost, vessel weight, and external footprint. A preliminary economic optimisation comparing insulation cost against long-term energy savings is outside the scope of this framework but is recommended for installations with high operating temperatures or extended continuous operation periods.

9.2 Rigorous Thermal Resistance Network

The rate of heat loss from the process vessel to the surrounding environment is governed by a series thermal resistance network spanning the internal fluid boundary layer, the vessel wall, the insulation layer, and the external boundary layer. Each resistance is evaluated using the finalised vessel geometry from Step 8.

The total thermal resistance of the vessel boundary is:

$$R_{total} = R_{internal} + R_{vessel\ wal} + R_{insulation} + R_{external}$$

9.2.1 Internal Convective Resistance

The internal convective resistance represents the thermal resistance of the boundary layer between the bulk process fluid and the inner vessel wall surface:

$$R_{internal} = \frac{1}{h_{internal} \times A_{total\ final}}$$

Where $h_{internal}$ is the natural convection coefficient of the process fluid at the vessel wall, estimated using the Churchill-Chu correlation applied to the internal vessel geometry. For well-mixed process fluids or those with low viscosity, $h_{internal}$ is typically high and $R_{internal}$ is the smallest resistance in the network, contributing minimally to total heat loss. For high-viscosity unagitated fluids, $h_{internal}$ may be meaningfully low and should be evaluated carefully.

For conservative preliminary estimation, $h_{internal}$ may be taken as 50 to 200 W/(m²·K) for low-to-moderate viscosity process fluids and 10 to 50 W/(m²·K) for high-viscosity organic fluids. These bounds should be replaced by a computed value using the Churchill-Chu correlation where the process fluid properties are well characterised.

9.2.2 Vessel Wall Conductive Resistance

The vessel wall conductive resistance represents the thermal resistance of the structural shell material. For a flat wall approximation applicable to large-diameter vessels with thin walls relative to vessel radius:

$$R_{vessel\ wall} = \frac{t_{wall}}{k_{wall} \times A_{total\ final}}$$

Where:

- t_{wall} is the vessel shell wall thickness (m)
- k_{wall} is the thermal conductivity of the vessel wall material (W/m·K)

For stainless steel TP316L vessel construction, $k_{wall} = 15.0$ W/(m·K). For carbon steel construction, $k_{wall} = 43.0$ W/(m·K). The vessel wall resistance is typically small relative to the insulation resistance and the external convective resistance but must not be omitted.

For vessels where the wall thickness is not negligible relative to the vessel radius — specifically smaller vessels below approximately 500mm internal diameter — the cylindrical wall conduction formula should be applied. The full cylindrical formulation is provided in Appendix A.

9.2.3 Insulation Conductive Resistance

The insulation layer presents the dominant thermal resistance in any properly insulated vessel. For the flat wall approximation:

$$R_{insulation} = \frac{L_{ins}}{k_{ins} \times A_{total\ final}}$$

Where:

- L_{ins} is the insulation layer thickness (m)
- k_{ins} is the thermal conductivity of the insulation material at operating temperature (W/m·K), upper bound value as specified in Section 9.1

For cylindrical vessel geometries where the insulation thickness is a significant fraction of the vessel radius, the cylindrical resistance formulation is more accurate and is provided in Appendix A.

9.2.4 External Convective Resistance

The external convective resistance represents the thermal resistance of the boundary layer between the outer insulation surface and the surrounding ambient air. It is governed by the external surface temperature of the insulation cladding and the ambient air movement conditions at the installation site:

$$R_{external} = \frac{1}{h_{external} \times A_{total\ final}}$$

Where $h_{external}$ is the external convective heat transfer coefficient (W/m²·K). Reference values for common installation conditions are:

External Condition	$h_{external}$ (W/m ² ·K)
Still indoor air — enclosed plant building	5 – 10
Gentle indoor air movement — ventilated building	10 – 15
Outdoor installation — sheltered from wind	15 – 25
Outdoor installation — exposed to prevailing wind	25 – 50

The upper bound of the applicable range is applied in all cases to maximise the estimated heat loss rate conservatively.

9.2.5 Total Resistance and Steady-State Heat Loss

The total thermal resistance is the sum of all series resistances:

$$R_{total} = R_{internal} + R_{vessel\ wal} + R_{insulation} + R_{external}$$

The worst-case steady-state environmental heat loss rate is determined by the worst-case thermal driving force across the total resistance network:

$$\Delta T_{loss} = T_{operating} - T_{worst\ ambient}$$

$$\Delta Q_{loss} = \frac{\Delta T_{loss}}{R_{total}}$$

Where $T_{worst\ ambient}$ is the historically coldest ambient temperature for heating applications and the historically hottest for cooling applications — the same reference temperature established in Step 1 and used consistently throughout the framework.

9.3 Q_{loss} Feedback Resolution

With Q_{loss} now rigorously calculated, it is formally compared against the updated placeholder value carried forward from Step 8:

$$Deviation = \frac{Q_{loss} - Q_{loss\ updated}}{Q_{loss\ updated}} \times 100\%$$

If the deviation is within ± 15 percent: The rigorously calculated Q_{loss} is consistent with the placeholder estimate within the conservatism embedded in the design factor of safety. The design proceeds without modification. Q_{loss} replaces $Q_{loss\ updated}$ as the definitive heat loss value for all subsequent calculations.

If Q_{loss} exceeds $Q_{loss\ updated}$ by more than 15 percent: The actual heat loss is substantially higher than the placeholder assumed. This means $Q_{balance\ calculated}$ in Step 1 is unconservative. The designer must return to Step 1, recalculate $Q_{balance}$ and Q_{design} using the rigorous Q_{loss} value, and re-run the framework from that point. This condition most commonly arises when the insulation tier selected in Step 1 was optimistic relative to the actual insulation specification confirmed here, or when the vessel surface area grew substantially during the displacement compensation in Step 8.

If Q_{loss} is more than 15 percent below $Q_{loss\ updated}$: The placeholder was overly conservative relative to the

actual insulation performance. The design is valid and may proceed. The designer may optionally revisit the thermal driver sizing to take advantage of the lower actual heat loss, reducing the required driver capacity and potentially achieving capital cost savings.

9.4 Total Thermal Driver Capacity

The thermal driver must simultaneously satisfy the peak process heat demands of all active vessels in the installation and counteract the steady-state environmental heat losses of all vessels under worst-case ambient conditions. The conservative design basis assumes all vessels are simultaneously at peak thermal demand — that is, all vessels are actively processing incoming cold batches concurrently. This represents the maximum instantaneous load the thermal driver will ever be required to meet and ensures the driver is never the bottleneck in a multi-vessel plant layout.

For an installation comprising n process vessels, the total required thermal driver capacity is:

$$Q_{total} = \sum_{i=1}^N (Q_{design,i} + Q_{loss,i})$$

Where:

- $Q_{design,i}$ is the fixed design thermal target for vessel i , from Step 1
- $Q_{loss,i}$ is the rigorously calculated steady-state heat loss for vessel i , from Section 9.2

For a single-vessel installation this simplifies to:

$$Q_{total} = Q_{design} + Q_{loss}$$

The designer should note that Q_{design} already contains the factor of safety multiplier applied in Step 1. The thermal driver is therefore sized with the FOS fully embedded in the process heat demand component, providing inherent resilience against long-term performance degradation and cold-start transients.

9.5 Thermal Driver Selection Range

The total thermal driver capacity Q_{total} establishes the upper bound of the required driver output — the maximum continuous thermal power the driver must be capable of delivering. The lower bound — the minimum continuous output the driver must sustain to prevent the process from cooling below its target temperature during low-demand periods — is governed solely by the standing heat losses:

$$Q_{driver\ max} = Q_{total}$$

$$Q_{driver\ min} = \sum_{i=1}^N Q_{loss,i}$$

These two values define the operational range within which the thermal driver must be capable of modulating its output. The ratio $Q_{driver\ min} / Q_{driver\ max}$ represents the required turndown ratio of the driver — the ability to reduce output from maximum to minimum without operational instability. A high turndown ratio — close to 1.0 — indicates that the standing losses are a large fraction of the total demand, which is characteristic of high-temperature installations with large vessel surface areas or poor insulation. A low turndown ratio — approaching 0 — indicates that standing losses are small relative to peak batch demand, which is characteristic of well-insulated systems processing large batch volumes.

The designer selects a commercial thermal driver — electric immersion heater assembly, fired thermal fluid heater, industrial heat pump, or other prime mover — whose rated capacity meets or exceeds $Q_{driver\ max}$ and whose minimum stable operating output is at or below $Q_{driver\ min}$. The driver's modulation capability — whether continuous via VFD or SCR control, or stepped via stage switching — must span the full range between $Q_{driver\ min}$ and $Q_{driver\ max}$ without instability.

For electric immersion heater configurations, the number of heating elements required to span this range is:

$$N_{max} = \frac{Q_{driver\ max}}{k}$$

$$N_{min} = \frac{Q_{driver\ min}}{k}$$

Where k is the rated thermal output of a single heating element (W). N_{max} elements are required at peak demand. N_{min} elements must remain active during minimum load holding to sustain the standing losses. The heating element rating k is determined by the designer based on the watt density limitations of the thermofluid and the element material, which are governed by the thermal battery design in the following step.

For fired thermal fluid heaters, the driver is selected from the manufacturer's standard capacity range at or above $Q_{driver\ max}$, with confirmation that the burner's minimum turndown output — typically 20 to 25 percent of rated capacity for modern modulating burners — is at or below $Q_{driver\ min}$.

For industrial heat pumps and vapor compression chillers used in cooling applications, the driver is selected with rated cooling capacity at or above $Q_{driver\ max}$, with confirmation that the compressor's minimum unloading step satisfies $Q_{driver\ min}$ without triggering anti-recycle protection.

9.6 Step 9 Output Summary

Upon completion of Step 9, the designer has established the following:

- The confirmed insulation material specification, thermal conductivity k_{ins} , and layer thickness L_{ins}
- The complete series thermal resistance network: $R_{internal}$, $R_{vessel\ wall}$, $R_{insulation}$, $R_{external}$ and R_{total}
- The rigorously calculated steady-state heat loss rate Q_{loss} for each vessel in the installation
- Resolution of the Q_{loss} feedback loop — confirmation that the design is valid or documentation of a return to Step 1 where the 15 percent deviation threshold has been exceeded
- The total thermal driver capacity Q_{total} and the standing loss floor $Q_{driver\ min}$
- The required driver turndown ratio and its implications for driver selection
- The selected commercial thermal driver specification — type, rated capacity, and modulation range — confirmed to span $Q_{driver\ min}$ to $Q_{driver\ max}$

These outputs define the thermal driver procurement specification and provide the thermal power boundary conditions that govern the thermal battery sizing in Step 10.

Step 10: Thermal Battery Sizing

Overview

Step 10 sizes and geometrically specifies the thermal battery — the utility fluid reservoir that stores conditioned thermofluid and supplies it continuously to the circulation loop serving the process vessels. The thermal battery is a custom-fabricated vessel whose volume is selected within a rigorously defined operating envelope bounded by a minimum and maximum constraint. Its geometry is derived using the same compound vessel methodology applied to the process vessel in Step 3.

The thermal battery performs two distinct engineering functions simultaneously. Thermodynamically, it acts as a buffer — absorbing the mismatch between the thermal driver's supply rate and the instantaneous demand of the process loops, preventing rapid temperature excursions that would destabilise the automation control system. Hydraulically, it acts as the fluid reservoir that keeps the closed circulation loop fully primed under all operating conditions, including transient startup sequences.

The volume of the thermal battery is the primary physical lever governing the stability and responsiveness of the entire automation system. A larger battery provides greater thermal inertia, smoother PID behaviour, and tolerance for aggressive batch loading — at the cost of higher capital expenditure, greater floor footprint, and longer thermal startup cycles. A smaller battery minimises cost and footprint but demands faster-acting control hardware and tighter automation tuning to prevent thermal runaway or depletion. The designer must make this tradeoff

explicitly and document the engineering basis for the selected volume.

The automation implications of the volume selection are addressed in Appendix D. The PLC implementation of the control loops that govern the thermal battery are detailed in Appendix E.

10.1 Mandatory Baseline — Hydronic Priming Volume

The hydronic priming volume $V_{hydronics}$ is the minimum fluid inventory required to completely fill all internal wetted passages of the closed circulation loop under operating pressures. It applies universally to every closed hydronic system regardless of the thermal driver type selected. It is evaluated first and establishes the absolute floor below which no thermal battery can be sized, irrespective of any other constraint.

The priming volume is the sum of three internal volume zones:

$$V_{hydronics} = V_{coil\ internal} + V_{distribution} + V_{ancillary}$$

Internal Coil Volume

The fluid volume contained within the bore of the heat exchange coil:

$$V_{coil\ internal} = \left(\frac{\pi d_{in}^2}{4} \right) \times L_{coil}$$

Where d_{in} is the internal pipe diameter of the selected NPS and L_{coil} is the optimized coil length from Step 7. Note that this calculation uses the internal pipe diameter — in contrast to the displacement calculation in Step 8 which used the external diameter. The distinction is critical: this quantity represents the thermofluid payload trapped inside the coil, not the volume the coil displaces in the process vessel.

Distribution Line Volume

The fluid volume residing within the external supply and return header lines connecting the thermal battery to the process vessel. For an installation with m discrete pipe runs of varying lengths l_i and internal diameters $d_{in,i}$:

$$V_{distribution} = \sum_{i=1}^m \left(\frac{\pi \times d_{in,i}^2}{4} \right) \times l_i$$

Where the pipe run lengths and diameters are determined from the plant layout. Where the detailed piping layout is not yet finalised, a conservative estimate of the distribution volume may be made based on the approximate distance between the thermal battery and the process vessel, using the same NPS selected for the coil circuit.

Ancillary Equipment Volume

The internal fluid hold-up of all secondary inline components — pump casing, manifold bodies, valve internals, inline strainers, and the minimum seal volume of the expansion vessel:

$$V_{ancillary} = V_{pump} + V_{manifold} + V_{expansion\ seal}$$

These values are obtained from manufacturer equipment datasheets. Where datasheets are not yet available, a conservative allowance of 5 to 10 percent of $V_{coil\ internal}$ is acceptable as a preliminary estimate.

The hydronic priming volume establishes the unconditional minimum:

$$V_{hydronic} \leq V_{battery} \text{ under all conditions}$$

If the thermal battery volume falls below $V_{hydronic}$ during initial system priming, the fluid level in the reservoir drops below the pump suction nozzle, breaking the fluid column and drawing air into the circulation loop. Air entrainment causes erratic pressure readings across inline sensors, triggers dry-run protection interlocks, and can destroy the pump impeller through cavitation within minutes of startup.

10.2 Driver-Specific Minimum Volume Constraints

In addition to the universal hydronic priming constraint, the selected thermal driver imposes its own minimum volume requirement governed by its specific mechanical and control limitations. Only the formulation corresponding to the driver type selected in Step 9 is evaluated. The result is combined with $V_{hydronic}$ to establish the overall minimum volume floor.

10.2.1 Electric Immersion Heater Arrays

For systems driven by electric resistance immersion elements, the minimum volume is governed by two concurrent requirements: element submersion safety and flux buffer capacity.

Element Submersion Volume $V_{clearance}$

The fluid level must fully submerge all active heating elements under all operating conditions, including the fluid contraction that occurs at low standby temperatures. If the fluid level drops and exposes any portion of an active element to the vapour space, the localised heat transfer coefficient collapses instantly. This triggers dry-firing burnout, mechanical deformation of the element sheath, and — in thermal oil systems — a severe fire or ignition hazard.

$V_{clearance}$ is determined from the physical geometry of the element bundle plus a safety overhead accounting for fluid contraction at the lowest expected standby temperature. This value is obtained from the element

manufacturer's installation requirements.

Flux Buffer Volume $V_{flux\ buffer}$

Electric elements operate at high localised surface heat flux — watt density. If the surrounding fluid mass is insufficient to absorb and distribute this flux before the element's internal high-temperature safety switches respond, the switches trip prematurely and the PID control loop begins oscillating. The minimum fluid mass required to damp these localised thermal spikes is:

$$V_{flux\ buffer} = \frac{Q_{element}}{\rho \times C_p \times \left(\frac{dT}{dt}\right)_{limit}}$$

Where $\left(\frac{dT}{dt}\right)_{limit}$ is the maximum allowable fluid temperature ramp rate that the control system can track without triggering safety faults (K/s), specified by the automation designer based on the SCR or VFD response capability.

The electric driver minimum volume is:

$$V_{electric} = V_{clearance} + V_{flux\ buffer}$$

10.2.2 Fired Thermal Fluid Heaters

For systems driven by gas, diesel, or biomass-fired thermal fluid heaters, the minimum volume is governed by the burner's physical turndown ratio. Fired burners cannot modulate below a minimum stable combustion output — typically 20 to 25 percent of rated capacity. If the fluid volume is too small, the fluid temperature rises through the entire controller deadband faster than the burner can throttle down, forcing the system into rapid short-cycling. In fired systems, short-cycling causes severe thermal fatigue in the heat exchanger tubes, accelerates carbon soot accumulation, and degrades the thermal oil through localised cracking.

$$V_{fired} = \frac{Q_{min\ burn} \times t_{min\ runtime}}{\rho \times C_p \times \Delta T_{deadband}}$$

Where:

- $Q_{min\ burn}$ is the minimum thermal output at the burner's lowest stable modulation point (W)
- $t_{min\ runtime}$ is the minimum continuous firing duration required to complete a clean combustion purge cycle (s)
- $\Delta T_{deadband}$ is the allowable control temperature deadband surrounding the system setpoint (K)

10.2.3 Vapor Compression Chillers

For cooling applications utilising vapor compression chillers, the minimum volume is governed by the compressor's anti-recycle timer — the mandatory restart delay enforced by the chiller controller to protect the motor windings from locked-rotor current damage:

$$V_{chiller} = \frac{Q_{min\ chiller} \times \tau_{anti\ recycle}}{\rho \times C_p \times \Delta T_{deadband}}$$

Where:

- $Q_{min\ chiller}$ is the chiller's cooling capacity at its lowest unloading step (W)
- $\tau_{anti\ recycle}$ is the compressor anti-recycle timer duration (s), typically 300 to 600 seconds for industrial refrigerant compressors
- $\Delta T_{deadband}$ is the control temperature deadband (K)

If the fluid volume is below $V_{chiller}$, a sudden heat spike following a compressor shutdown will deplete the cold thermal battery before the anti-recycle timer expires, causing an uncontrolled temperature overshoot in the process vessel.

10.2.4 Overall Minimum Volume

The overall minimum volume floor is the maximum of the hydronic priming constraint and the applicable driver-specific constraint:

$$V_{min} = \max(V_{hydronic}, V_{driver\ specific})$$

Where $V_{driver\ specific}$ is whichever of $V_{electric}$, V_{fired} , or $V_{chiller}$ applies to the installation. This formulation ensures that both the hydraulic integrity of the loop and the mechanical stability of the thermal driver are simultaneously protected.

10.3 Maximum Volume Constraint

The maximum thermal battery volume is the largest reservoir that is economically and operationally justifiable for the installation. It is defined as the volume required to sustain peak process demand through one complete batch cycle solely from stored thermal energy — with the thermal driver assumed to have failed — without the fluid temperature dropping below the minimum allowable operating limit.

This worst-case autonomy analysis provides the absolute upper bound on useful thermal storage. A reservoir larger than V_{max} provides thermal inertia beyond what a single batch cycle can consume, which translates to unnecessary

capital cost, excessive floor footprint, prolonged thermal startup cycles, and elevated standing heat losses.

The allowable temperature drawdown of the thermofluid during autonomous batch operation is:

$$\Delta T_{allow} = T_{operating} - T_{min\ allowed}$$

Where $T_{min\ allowed}$ is the lowest thermofluid temperature at which the circulation loop can continue to deliver adequate heat transfer to the process vessel. For heating applications this is typically the process fluid target temperature plus a minimum driving force margin of 5 to 10°C.

The maximum volume is derived from the first-law energy balance across the thermal battery during a full peak-load batch cycle with no driver input:

$$V_{max} = \frac{Q_{peak}}{\rho \times C_p \times \Delta T_{allow} \times \eta_{usable}}$$

Where:

- Q_{peak} is the peak thermal power demand of the installation (W), equal to Q_{total} from Step 9
- ρ and C_p are the thermofluid density and specific heat capacity at bulk operating temperature
- ΔT_{allow} is the allowable temperature drawdown as defined above (K)
- η_{usable} is the usable thermal fraction of the reservoir, accounting for internal thermal stratification and parasitic standing losses. A conservative value of 0.80 is applied as the default

The full first-law derivation proving the V_{max} autonomy boundary is provided in Appendix A.

10.4 System Stability Assessment

The thermal stability of the system at any candidate volume within the operating envelope is quantified by the thermal time constant S_t :

$$S_t = \frac{\rho \times V_{battery} \times C_p}{Q_{peak}}$$

Where S_t has units of seconds and represents the time the thermal battery can sustain peak demand before its temperature drops by an amount equivalent to the full peak thermal power acting on the fluid mass. Higher values of S_t indicate greater thermal inertia and smoother, more stable automation behaviour. Lower values indicate a highly reactive system that demands fast-acting control hardware to prevent temperature excursions.

The following reference bands provide practical guidance on control hardware requirements as a function of S_t :

S _t Range	System Behaviour	Control Hardware Implication
S _t > 1800 s (30 min)	High inertia — very stable	Standard PID with moderate tuning. Conventional VFD and SSR control adequate.
600 s < S _t ≤ 1800 s	Moderate inertia — stable under normal loading	Careful PID tuning required. VFD with responsive ramp rates recommended.
180 s < S _t ≤ 600 s	Low inertia — reactive	Fast-acting SCR power controllers are mandatory. Predictive feedforward PID recommended.
S _t ≤ 180 s	Very low inertia — highly volatile	Advanced model-predictive control required. Standard PID will not stabilize the loop reliably.

The designer evaluates S_t at both V_{min} and V_{max} to understand the full stability range of the system, then selects a target volume that delivers an S_t consistent with the control hardware available for the installation.

10.5 Volume Selection

The designer selects the target thermal battery volume $V_{battery}$ within the operating envelope:

$$V_{min} \leq V_{battery} \leq V_{max}$$

The selection is governed by three primary factors which the designer must weigh against each other based on the specific constraints of the installation:

Available Capital — A larger battery requires a larger custom vessel, more thermofluid inventory, and a larger installation footprint. Where capital is constrained, selecting $V_{battery}$ closer to V_{min} minimises upfront cost but transfers that cost to more sophisticated control hardware and tighter commissioning effort.

Plant Floor Footprint — Where floor space is limited, $V_{battery}$ is driven toward V_{min} . The resulting low thermal inertia must be compensated by fast-acting automation hardware as indicated by the S_t assessment in Section 10.4.

Control Hardware Capability — Where the available control hardware is conventional — standard VFD and PID — the designer must select $V_{battery}$ high enough to achieve $S_t > 600$ seconds. Where advanced SCR controllers and model-predictive algorithms are available, $V_{battery}$ can approach V_{min} without sacrificing loop stability.

The selected $V_{battery}$ and its corresponding S_t values are recorded as the primary outputs of the volume selection decision.

10.6 Thermal Battery Tank Geometric Specification

With $V_{battery}$ confirmed, the custom thermal battery tank is geometrically dimensioned using the same compound vessel methodology applied in Step 3. The same cap profile selected for the process vessel is applied here unless specific operational requirements — such as gravity drainage of the thermofluid for maintenance — dictate otherwise.

The same aspect ratio convention is applied:

$$h_{1\ battery} = 4 \times h_{2\ battery}$$

Setting the relevant volume formula equal to $V_{battery}$ and solving for $h_{2\ battery}$ and subsequently $d_{battery}$, $h_{1\ battery}$, and $h_{total\ battery}$ using the algebraic procedure established in Step 3 and detailed in Appendix A.

The same aspect ratio sanity checks from Step 3 Section 3.2.1 apply:

- Minimum internal diameter of 400mm — thermal battery tanks are typically smaller than process vessels and this lower bound reflects the minimum practical fabrication dimension for a cylindrical pressure vessel with internal heating elements or heat exchanger connections
- Height-to-diameter ratio between 0.8 and 4.0

Where the geometry falls outside these bounds, the aspect ratio is adjusted within a reasonable range to recover a practicable geometry, with the modification documented. The thermal battery tank geometric specification comprises:

- Internal diameter: $d_{battery}$
- Cylindrical shell height: $h_{1\ battery}$
- Cap height: $h_{2\ battery}$
- Total height: $h_{total\ battery}$
- Cap profile: as selected
- Internal volume: $V_{battery}$

This specification is the complete buildable geometry of the thermal battery tank and is carried forward to the detailed structural and fabrication engineering stage.

10.7 Step 10 Output Summary

Upon completion of Step 10, the designer has established the following:

- The hydronic priming volume $V_{hydronic}$ with all three components — coil internal volume, distribution line volume, and ancillary equipment volume — individually quantified
- The driver-specific minimum volume constraint — $V_{electric}$, V_{fired} or $V_{chiller}$ — evaluated for the selected driver type
- The overall minimum volume floor V_{min}
- The maximum volume boundary V_{max} with the allowable temperature drawdown ΔT_{allow} and usable thermal fraction η_{usable} documented
- The thermal time constant S_t evaluated at V_{min} and V_{max} with the corresponding control hardware implications identified
- The selected thermal battery volume $V_{battery}$ with the engineering basis for the selection documented against the three decision factors in Section 10.5
- The complete buildable geometric specification of the custom thermal battery tank: $d_{battery}$, $h_{1 battery}$, $h_{2 battery}$, $h_{total battery}$

Together with the outputs of all preceding steps, the thermal battery specification completes the primary engineering design of the VCH hydronic system. The process vessel geometry, internal coil assembly, circulation pump, thermal driver, and thermal battery are all fully defined as buildable specifications ready for detailed structural, electrical, and automation engineering.

Step 11: Design Verification and Release

Overview

Step 11 is the formal verification gate of the VCH Sizing Framework. It consolidates every governing constraint established across the preceding ten steps into a single structured audit, confirming that the complete design is internally consistent, physically realisable, and thermally adequate before it is released for detailed structural, electrical, and automation engineering.

No new calculations are introduced in this step. All values referenced here have been computed in earlier steps and are being re-examined collectively for the first time. The purpose of this collective examination is to catch inconsistencies, accumulated rounding errors, and constraint violations that may not have been apparent when each step was evaluated in isolation.

The verification is organised into four domains — thermal, hydraulic, geometric, and control stability — followed by two cross-cutting global checks that span the entire framework. Each check states the acceptance criterion explicitly. Where a check fails, the corrective action identifies the step to which the designer must return before the design can be released.

The design is released for detailed engineering only when every check in this step is documented as passed. A design that proceeds to detailed engineering with unresolved verification failures carries a significant risk of commissioning failure, costly rework, and operational underperformance.

11.1 Thermal Verification Checks

T1 — Design Thermal Target Consistency

Verify that Q_{design} carried forward through all steps is identical to the value established in Step 1 and has not been modified, re-derived, or approximated at any intermediate step.

Acceptance criterion: $Q_{design} (Step\ 1) = Q_{design} (in\ all\ other\ steps)$

Corrective action if failed: Identify where Q_{design} was modified then restore the Step 1 value. Re-run all affected steps.

T2 — Batch Energy Component Completeness

Verify that Q_{batch} accounts for all applicable energy components for the specific process fluid — sensible heating of the solid phase, latent heat of fusion where applicable, and sensible heating of the liquid phase. Confirm that $T_{worst_ambient}$ was used as the incoming material temperature baseline.

Acceptance criterion: $Q_{total} = Q_{solid} + Q_{melt} + Q_{liquid}$, with $Q_{melt} = 0$, only where the process fluid undergoes no phase transition within the operating range.

Corrective action if failed: Return to Step 1 and recompute Q_{batch} with all applicable components included.

T3 — Thermofluid Supply Temperature Adequacy

Verify that the thermofluid supply temperature T_{supply} exceeds the process fluid target temperature $T_{operating}$ by a meaningful margin. Heat transfer from the coil to the process fluid requires a positive driving force at every point along the coil length. If T_{supply} approaches $T_{operating}$, ΔT_{drive} collapses and the heat transfer rate becomes negligible regardless of coil surface area.

Acceptance criterion: $T_{supply} - T_{operating} \geq 10^{\circ}C$

Corrective action if failed: Return to Step 2 and select a thermofluid with a higher operating temperature ceiling, or increase the thermal driver supply temperature within the thermofluid's rated stability limit.

T4 — ΔT_{coil} Constraint Compliance

Verify that the thermofluid temperature differential across the coil inlet and outlet satisfies the framework's property stability constraint.

Acceptance criterion: $\Delta T_{coil} \leq 0.10 \times (T_{operating} - T_{worst\ ambient})$

Corrective action if failed: Return to Step 4 and recalculate Q_{flow} with a reduced ΔT_{coil} . Re-run Steps 4 through 10.

T5 — Steady-State Thermal Adequacy

Verify that the maximum transmissible thermal power at steady-state operating conditions meets or exceeds Q_{design} .

Acceptance criterion: $Q_{max\ steady} \geq Q_{design}$

Corrective action if failed: Return to Step 6 and identify the binding thermal resistance. Return to Step 4 if the coil surface area is insufficient and increment NPS.

T6 — Cold-Start Thermal Adequacy

Verify that the maximum transmissible thermal power under worst-case cold-start viscosity conditions meets or exceeds Q_{design} . This is the more demanding of the two thermal adequacy checks and governs coil surface area sizing.

Acceptance criterion: $Q_{extreme\ temp} \geq Q_{design}$

Corrective action if failed: Return to Step 6. Review U_{cold} calculation and confirm worst-case viscosity was correctly applied. If the coil surface area is the bottleneck, return to Step 4 and increment NPS.

T7 — Heat Exchanger Effectiveness Bound

Verify that the target heat exchanger effectiveness does not approach or exceed physically realisable limits.

Acceptance criterion: $0 < \varepsilon_{target} < 0.9$

Corrective action if failed: If $\varepsilon_{target} \geq 0.90$, the thermofluid supply temperature is insufficient relative to the required heat transfer rate. Return to Step 2 and increase T_{supply} or select a thermofluid with better thermophysical properties at the operating condition.

T8 — Q_{loss} Feedback Resolution Confirmation

Verify that the Q_{loss} feedback loop has been formally resolved in Step 9 and that the rigorous Q_{loss} value is within 15 percent of the Step 8 updated placeholder. Confirm that if the deviation exceeded 15 percent, the design was returned to Step 1 and Q_{design} was recalculated before proceeding.

Acceptance criterion: $\frac{|Q_{loss\ rigorous} - Q_{loss\ placeholder}|}{Q_{loss\ placeholder}} \leq 15\%$, or documented evidence that the design loop return to Step 1 was executed.

Corrective action if failed: Return to Step 9, verify the thermal resistance network calculation, and if the deviation threshold is exceeded, return to Step 1.

T9 — Thermal Driver Capacity Adequacy

Verify that the selected commercial thermal driver's rated output meets or exceeds the total thermal demand of the installation under peak simultaneous loading conditions.

Acceptance criterion: $Q_{driver\ rated} \geq Q_{total} \Sigma(Q_{design,i} + Q_{loss,i})$

Corrective action if failed: Return to Step 9 and select a higher-capacity thermal driver or reduce Q_{total} by improving vessel insulation.

11.2 Hydraulic Verification Checks

H1 — Velocity Ceiling Compliance

Verify that the confirmed steady-state operating fluid velocity within the coil does not exceed the velocity ceiling assigned in Step 4 for the confirmed material-fluid pairing.

Acceptance criterion: $v_{operating} \leq v_{max}$

Corrective action if failed: Return to Step 5 and adjust the VFD clamping frequency. If VFD adjustment cannot resolve the violation, return to Step 4 and increment NPS.

H2 — Reynolds Number and Flow Regime Confirmation

Verify that the flow regime classification used in the Nusselt number correlation selection in Step 6 is consistent with the Reynolds number at the confirmed operating flow rate.

Acceptance criterion: If $Re > 10,000$ — Gnielinski correlation was applied. If $Re \leq 10,000$ — Sieder-Tate correlation was applied.

Corrective action if failed: Return to Step 6 and recompute h_{in} using the correct correlation for the confirmed Re . Recompute U , Q_{max} , and all dependent outputs.

H3 — Dean Number Regime Consistency

Verify that the helical friction factor correlation applied in Step 5 is consistent with the Dean number at the confirmed operating condition.

Acceptance criterion: If $De < 13.5$ — straight pipe friction factor applied. If $13.5 \leq De \leq 2000$ — laminar Ito correlation applied. If $De > 2000$ — turbulent Ito correlation applied.

Corrective action if failed: Return to Step 5 and recompute $f_{helical}$ using the correct Ito correlation. Rebuild the system curve and re-identify the duty point.

H4 — Pump NPSH Margin

Verify that the available Net Positive Suction Head of the installation exceeds the pump's required NPSH at the confirmed operating flow rate by the specified minimum margin.

Acceptance criterion: $NPSH_{available} - NPSH_{required} \geq 0.5m$

Corrective action if failed: Return to Step 5. Review the pump inlet conditions — increase the fluid head above the

pump suction nozzle, reduce the suction line resistance, or select a pump with lower $NPSH_{required}$.

H5 — System Pressure Drop Within Pump Seal Limits

Verify that the total system pressure drop at the confirmed operating flow rate does not exceed the rated operating pressure of the pump mechanical seal assembly.

Acceptance criterion: $\Delta P_{system} \leq \Delta P_{seal rated}$

Corrective action if failed: Return to Step 5. Consider splitting the single coil circuit into two parallel circuits to reduce the pressure drop presented to each pump, and re-evaluate the system curve.

H6 — Minor Loss Allowance Adequacy

Verify that the minor loss contribution applied in Step 5 was based on either an itemised K-factor schedule or the conservative 20 percent blanket allowance, and that this allowance has been flagged for replacement by an itemised calculation once the detailed piping layout is finalised.

Acceptance criterion: Minor losses were included in the system curve construction and the method applied is documented.

Corrective action if failed: Return to Step 5 and recompute the system curve with minor losses included.

H7 — Hydronic Priming Volume Satisfied

Verify that the selected thermal battery volume $V_{battery}$ is at or above the hydronic priming volume $V_{hydronic}$ under all operating conditions including the minimum standby fluid level.

Acceptance criterion: $V_{battery} \geq V_{hydronic}$

Corrective action if failed: Return to Step 10 and increase $V_{battery}$ to at least $V_{hydronic}$.

11.3 Geometric Verification Checks

G1 — Process Vessel Volume Consistency

Verify that the gross fabrication volume V_{gross} computed from the finalised vessel dimensions equals V_{target} plus $V_{displacement}$ within an acceptable rounding tolerance.

$$\text{Acceptance criterion: } \frac{|V_{gross \text{ computed}} - (V_{target} + V_{displacement})|}{V_{target}} \leq 1\%$$

Corrective action if failed: Recompute the vessel dimensions from V_{gross} using the Step 3 volume formula and update the dimensional specification in Step 8.

G2 — Coil Radial Clearance

Verify that the radial clearance between the outer edge of the coil and the internal vessel wall meets the minimum requirement on all sides, including after any vessel diameter adjustment made in Step 8.

$$\text{Acceptance criterion: } \frac{d_{tank \text{ final}} - d_{coil}}{2} - \frac{d_{pipe}}{2} \geq 0.100m$$

Corrective action if failed: Return to Step 4 and confirm d_{coil} was correctly computed from d_{tank} . If Step 8 reduced d_{tank} below the Step 4 basis, the coil diameter must be revisited — note this is the one geometric exception where a minor coil parameter adjustment is permissible to satisfy a safety clearance requirement.

G3 — Coil Pitch Within Bounds

Verify that the optimised helical pitch p_{coil} satisfies both the minimum and maximum pitch bounds established in Step 7.

$$\text{Acceptance criterion: } p_{min} \leq p_{coil} \leq \frac{h_1}{3}$$

Corrective action if failed: Return to Step 7 and apply the corrective actions specified in Sections 7.4.1 and 7.4.2.

G4 — Coil End Clearance

Verify that the clearance between the top and bottom of the wound coil and the adjacent cap regions meets the minimum requirement.

Acceptance criterion: $h_{clearance} \geq 50\text{mm at each end}$

Corrective action if failed: Return to Step 7 and reduce N_{coil} by one turn. Recheck thermal adequacy with the reduced surface area.

G5 — Optimised Coil Within Practical Boundary

Verify that the optimised coil length does not exceed the maximum practical length established in Step 4.

Acceptance criterion: $L_{coil} \leq L_{practical}$

Corrective action if failed: Return to Step 4 and increment NPS. The current pipe diameter cannot provide sufficient surface area within the vessel envelope.

G6 — Vessel Aspect Ratio Compliance

Verify that the finalised process vessel and thermal battery tank both satisfy the practical aspect ratio bounds.

Acceptance criterion: $0.8 \leq \frac{h_{final}}{d_{tank\ final}} \leq 4.0$ for process vessel. $0.8 \leq \frac{h_{total\ battery}}{d_{battery}} \leq 4.0$ for thermal battery tank.

Corrective action if failed: Return to Step 3 or Step 10 respectively and adjust the aspect ratio within the permissible range.

G7 — Displacement Calculation Uses External Diameter

Verify that the coil displacement volume was computed using the external pipe diameter d_{pipe} and not the internal diameter d_{in} .

Acceptance criterion: $V_{coil} = \left(\frac{\pi d_{pipe}^2}{4} \right) \times L_{coil}$

Corrective action if failed: Return to Step 8 and recompute $V_{displacement}$ using d_{pipe} . Update V_{gross} and the vessel dimensional specification accordingly.

11.4 Control Stability Verification Checks

C1 — Thermal Battery Volume Within Operating Envelope

Verify that the selected thermal battery volume lies within the bounds defined in Step 10.

Acceptance criterion: $V_{min} \leq V_{battery} \leq V_{max}$

Corrective action if failed: Return to Step 10 and select a volume within the defined envelope.

C2 — Thermal Time Constant and Control Hardware Consistency

Verify that the thermal time constant S_t at the selected $V_{battery}$ is consistent with the control hardware specified for the installation. Where S_t falls in the low or very low inertia bands, confirm that the corresponding fast-acting control hardware has been specified.

Acceptance criterion: S_t and control hardware specification are consistent per the reference table in Step 10 Section 10.4.

Corrective action if failed: Either increase $V_{battery}$ to achieve a higher S_t consistent with the available control hardware, or upgrade the control hardware specification to match the low-inertia demands of the selected volume.

C3 — Driver Turndown Ratio Achievable

Verify that the selected commercial thermal driver is capable of modulating its output across the full range from $Q_{driver\ min}$ to $Q_{driver\ max}$ without operational instability.

Acceptance criterion: *Driver minimum stable output* $\leq Q_{driver\ min}$ and
driver rated output $\geq Q_{driver\ max}$

Corrective action if failed: Return to Step 9 and select a driver with a wider modulation range, or adjust the insulation specification to reduce $Q_{driver\ min}$.

11.5 Global Cross-Cutting Checks

GL1 — Global Energy Balance

Verify that the total thermal driver capacity is consistent with the sum of all individual vessel design thermal targets and rigorous heat loss values across the complete installation.

$$Q_{driver\ rated} \geq \sum_{i=1}^n (Q_{design,i} + Q_{loss,i})$$

This check catches arithmetic inconsistencies that may have accumulated across multiple steps — for example, a Q_{loss} value updated in Step 9 that was not propagated to the Step 9 summation, or a Q_{design} value that differs between Steps 1 and 9 due to an undocumented intermediate modification.

Acceptance criterion: The inequality is satisfied and all component values can be traced directly to their source step without modification.

Corrective action if failed: Trace each term back to its source step, identify the inconsistency, correct it, and re-run all affected downstream steps.

GL2 — Worst-Case Ambient Temperature Consistency

Verify that the same worst-case ambient temperature $T_{worst\ ambient}$ has been applied consistently across all steps where it appears as a boundary condition — specifically Steps 1, 5, 9, and 10.

Acceptance criterion: $T_{worst\ ambient}$ used in $\Delta T_{coil}(\text{step 1}) = T_{worst\ ambient}$ used in $Q_{loss\ calculation}(\text{step 9}) = T_{worst\ ambient}$ used in V_{max} derivation (Step 10)

Corrective action if failed: Identify which step used a different ambient reference, correct the inconsistency, and re-run all affected steps.

GL3 — Thermofluid Property Evaluation Temperature Consistency

Verify that thermofluid thermophysical properties have been evaluated at a consistent bulk mean temperature throughout the framework, and that the worst-case cold-start viscosity has been applied only where specified — specifically in the U_{cold} calculation in Step 6.

Acceptance criterion: ρ , C_p , k_{fluid} used in Steps 4, 5, and 10 are all evaluated at the same T_{bulk} . μ_{cold} used in Step 6 U_{cold} calculation is the cold-start viscosity, not the bulk operating viscosity.

Corrective action if failed: Identify the step where an inconsistent property value was used, correct it, and recompute all affected outputs.

GL4 — Unit System Consistency

Verify that all calculations have been performed in a consistent unit system throughout the framework. All governing equations in this framework are expressed in SI units — metres, kilograms, seconds, watts, and kelvin. Any inputs provided in non-SI units must have been converted before use.

Acceptance criterion: All dimensional quantities can be traced to SI unit inputs. No mixed-unit calculations are present.

Corrective action if failed: Identify the unit inconsistency, convert to SI, and recompute all affected outputs from that point forward.

11.6 Verification Summary Record

Upon completing all checks, the designer compiles a verification summary record documenting the outcome of each check. The record must include for each check: the check identifier, the acceptance criterion, the computed value against which it was assessed, the pass or fail outcome, and — where applicable — the corrective action taken and the step re-run as a result.

The verification summary record is a mandatory deliverable of the VCH design process. It constitutes the auditable evidence that the design has been formally verified and provides the traceability required for engineering sign-off, client review, and regulatory submission where applicable.

11.7 Design Release Statement

When all checks in Sections 11.1 through 11.5 are documented as passed and the verification summary record is complete, the VCH design is formally released for detailed engineering.

The released design package comprises the following primary deliverables, each traceable to its source step:

- Process vessel buildable specification — geometry, volume, cap profile, and material (Steps 3 and 8)
- Internal coil assembly specification — L_{coil} , d_{coil} , d_{pipe} , $NPS\ schedule$, p_{coil} , N_{coil} , h_{coil} (Steps 4 and 7)
- Circulation pump specification — model, performance curve, VFD frequency range, seal specification (Step 5)
- Overall heat transfer coefficient documentation — U_{steady} and U_{cold} with all resistance components itemised (Step 6)
- Thermal driver specification — type, rated capacity, modulation range (Step 9)
- Thermal battery tank buildable specification — geometry, volume, and thermofluid inventory (Step 10)
- Verification summary record — all checks passed and documented (Step 11)

This package provides the complete primary engineering basis from which structural detailing, electrical sizing, instrumentation selection, and automation programming proceed.

References and Core Standards

The Velocity-Constrained Hydronic (VCH) Sizing Framework was developed as an original, integrated layout methodology. However, the foundational thermodynamic equations, fluid mechanics, and material dimensional constraints driving its internal calculation loops are firmly anchored in the following standardized texts and industrial codes:

Commercial Manufacturing and Material Standards 5. ASME B36.10M-2022. *Welded and Seamless Wrought Steel Pipe*. The American Society of Mechanical Engineers. (Governing standard for standardized external diameters and wall thicknesses utilized in the commercial NPS selection loop). 6. ASME B36.19M-2022. *Stainless Steel Pipe*. The American Society of Mechanical Engineers. (Governing standard for austenitic and duplex stainless steel commercial dimensional limits).

Industrial Process Plant Design 7. Green, D. W., & Southard, M. Z. (2018). *Perry's Chemical Engineers' Handbook* (9th ed.). McGraw-Hill Education. (Reference for empirical boundary assumptions, organic fluid fouling factors, and high-viscosity product handling).

[1] F. P. Incropera, D. P. DeWitt, T. L. Bergman, and A. S. Lavine, *Fundamentals of Heat and Mass Transfer*, 7th ed. Hoboken, NJ: John Wiley & Sons, 2011. (Provides the foundational equations for external natural convection and overall heat transfer coefficients used in Steps 3 and 6).

[2] H. Ito, "Friction factors for turbulent flow in curved pipes," *Journal of Basic Engineering*, vol. 81, no. 2, pp. 123-132, 1959. (The original engineering source for the Ito friction factor correlation used to determine hydraulic resistance in helical coils).

[3] E. F. Schmidt, "Heat transfer and pressure drop in coiled tubes," *Chemical Engineering Technology*, vol. 39, no. 11, pp. 781-789, 1967. (The empirical source for the helical convective heat transfer correlations applied during your iterative tuning loops).

[4] W. M. Rohsenow, J. P. Hartnett, and Y. I. Cho, *Handbook of Heat Transfer*, 3rd ed. New York, NY: McGraw-Hill, 1998. (Serves as the reference baseline for the thermophysical property datasets of hydronic process fluids and glycol concentrations listed in Appendix B).

[5] K. J. Åström and T. Hägglund, *Advanced PID Control*, Research Triangle Park, NC: ISA (Instrumentation, Systems, and Automation Society), 2006. (The automation baseline utilized for configuring the discrete-time positional PID loops and integral anti-windup clamping logic detailed in Appendix E).